

NEW SUMMIT COLLEGE

(Affiliated to Tribhuvan University)



Lab Report of
Computer Networking
CACS 303

Bachelors of Computer Application
Faculty of Humanities and Social Sciences

Submitted by:

Name: Sudha Pantha

Semester: V

Program: BCA

Submitted to:

Name: Apil Adhikari



Lab Program Number: ...09...

Date: 2080-10-08

Title: To configure DHCP and Provide Dynamic Ip to the nodes in network using cisco packet tracer.

Theory:

Dynamic Host Configuration Protocol (DHCP) is a network management protocol used to automate the process of configuring devices on IP networks, thus allowing them to use network services such as DNS, NTP, and any communication protocol based on UDP or TCP.

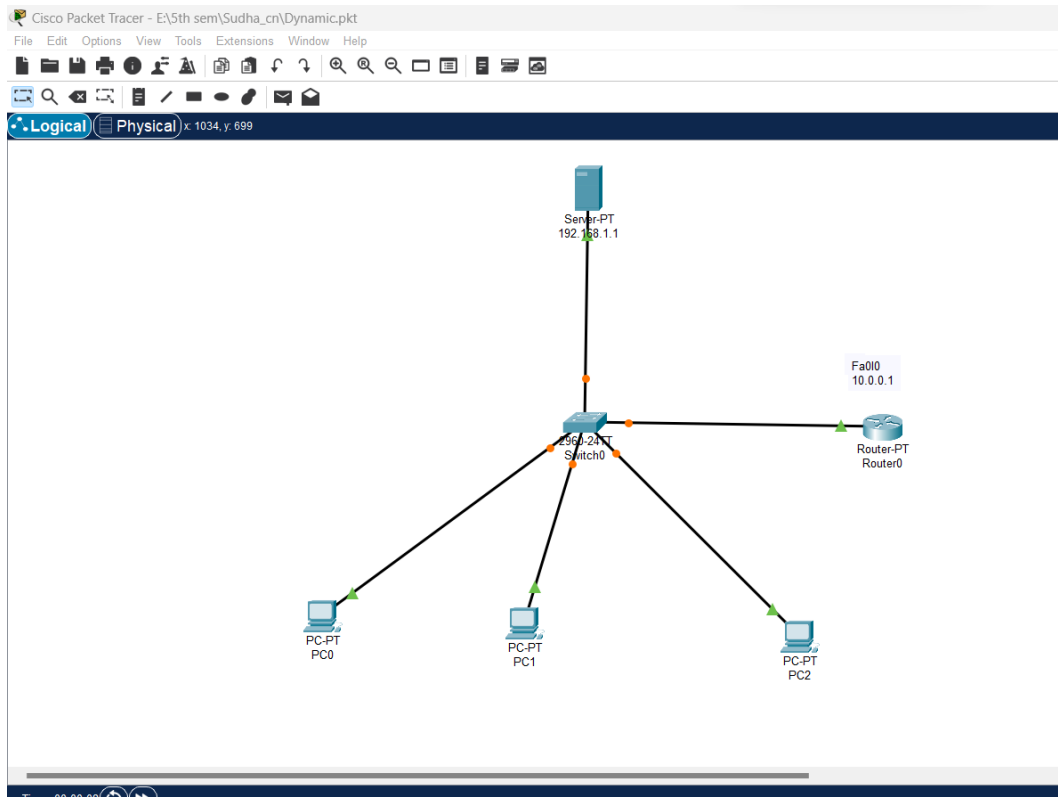
Procedure:

Step 1: Open Cisco Packet Tracer desktop and select all the devices given below:

S.No	Device	Model Name	Qty.
01	PC	PC	03
02	Switch	PT-Switch	01
03	Server	Server-PT	01
04	Router	Router-PT	01

- In the “Devices” tab on the left, select “End Devices” and drag a “Generic” Pc or laptop to the workspace.
- Similarly, drag a “Switch”, “Server” and “Router” to the workspace.
- Click on the “Connections” icon on the left-hand side of the screen.
- Choose the appropriate cable type usually, Ethernet connections use straight-through cables.
- Click on one device’s interface port and then click on the other devices interface to establish a connection.
- Then, create a network topology as shown below the image.

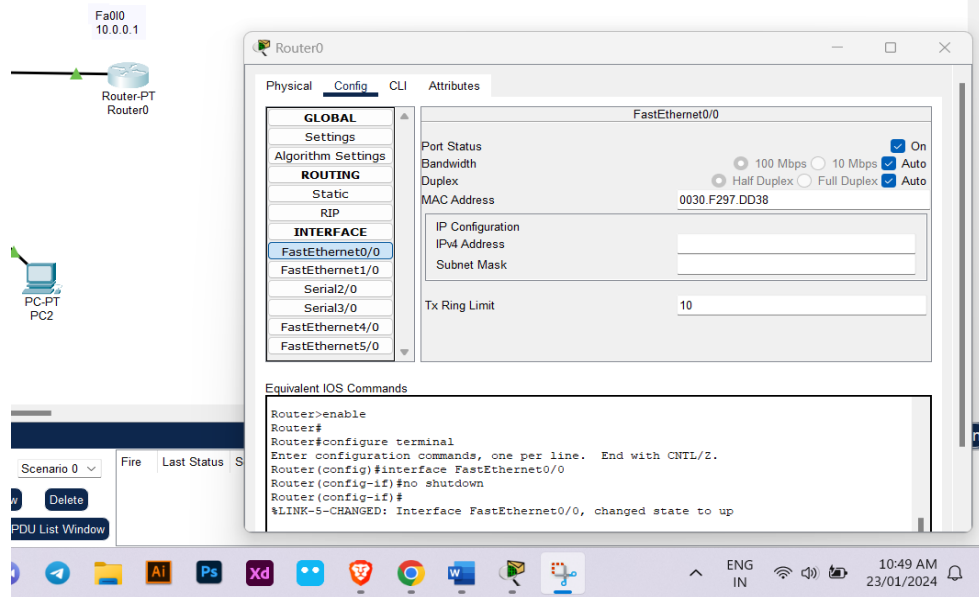
Name: Sudha Pantha



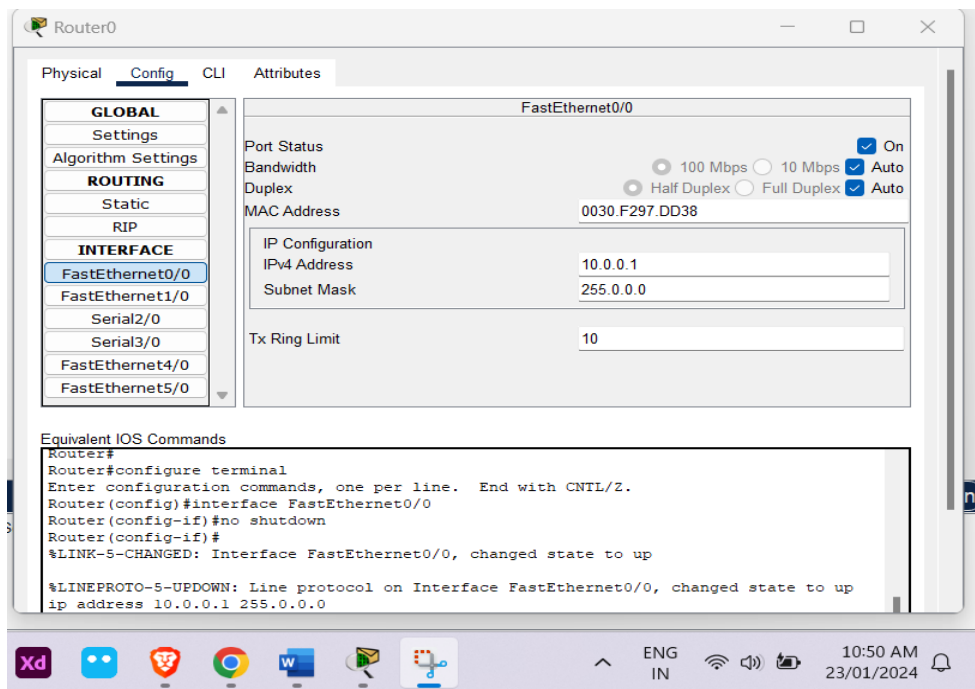
Step 2: Assigning Ip to the Router

- Click on the router in your topology.
- Go to the config and select FastEthernet0/0.
- Click on Port status on.

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- Then, Assign Ipv4 address to the Router.



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Step 3: Assigning Ip address to the Server

- Click on the server of your topology
- Go to the Ip Config and assign static Ip to the server.

10.0.0.10

Physical Config Services **Desktop** Programming Attributes

IP Configuration X

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.0.0.10

Subnet Mask 255.0.0.0

Default Gateway 10.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::290:21FF:FEA9:5EB6

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Ps Xd 11:14 AM 23/01/2024

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- Then go to the services, click on the DHCP and turn on the service.

The screenshot shows a network configuration window titled '10.0.0.10'. The 'Services' tab is selected, and the 'DHCP' service is turned on. The configuration details for the 'serverPool' are as follows:

Interface	Service
FastEthernet0	On

Pool Name: serverPool

Default Gateway: 10.0.0.1

DNS Server: 0.0.0.0

Start IP Address: 10.0.0.10

Subnet Mask: 255.0.0.0

Maximum Number of Users: 512

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Buttons: Add, Save, Remove

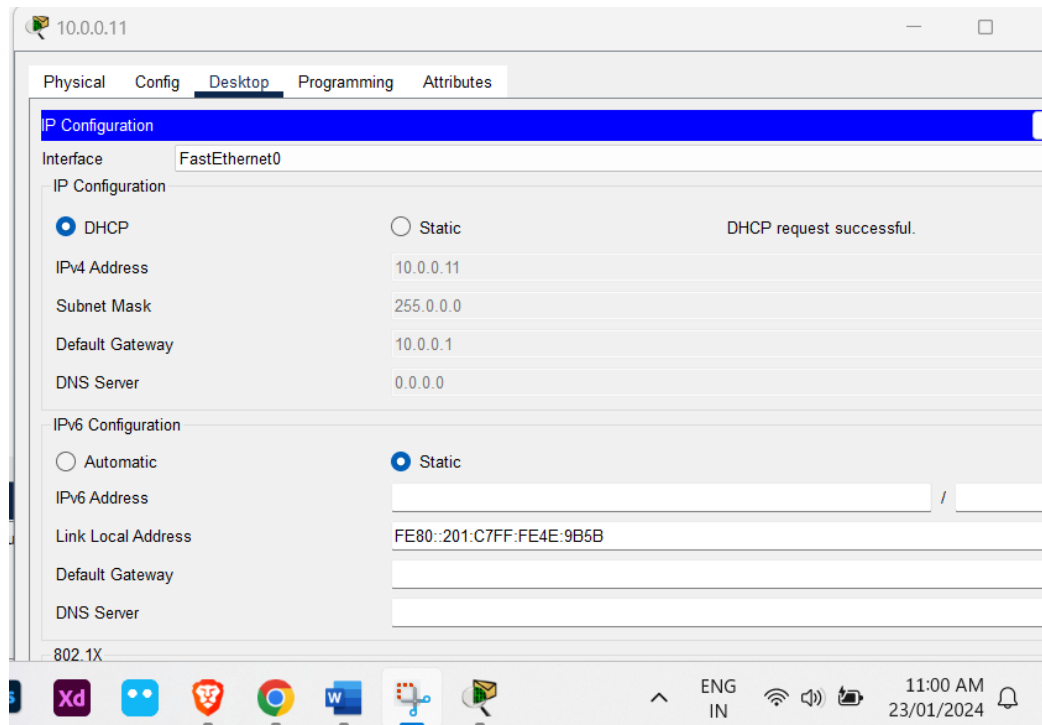
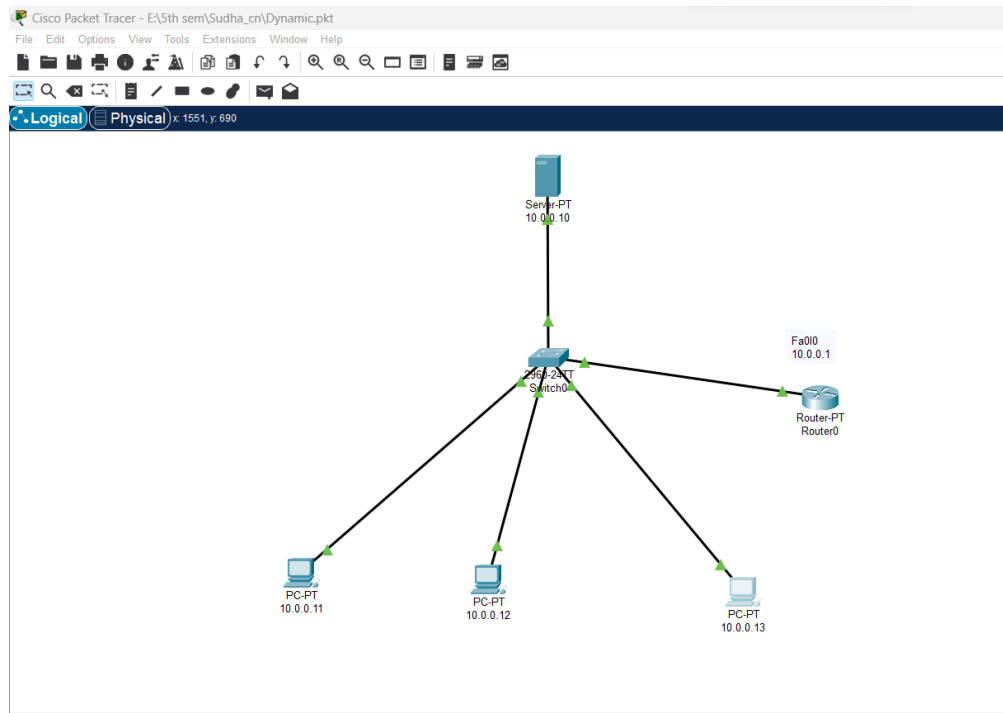
Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	10.0.0.1	0.0.0.0	10.0.0.10	255.0.0.0	512	0.0.0.0	0.0.0.0

Simulation

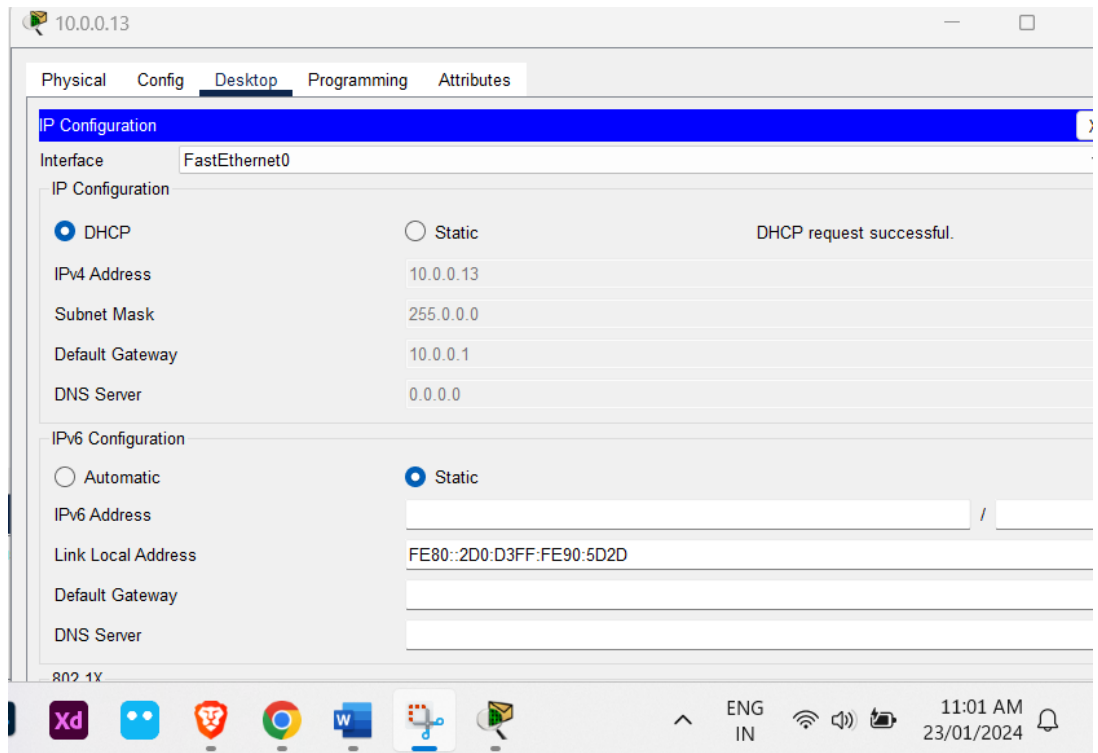
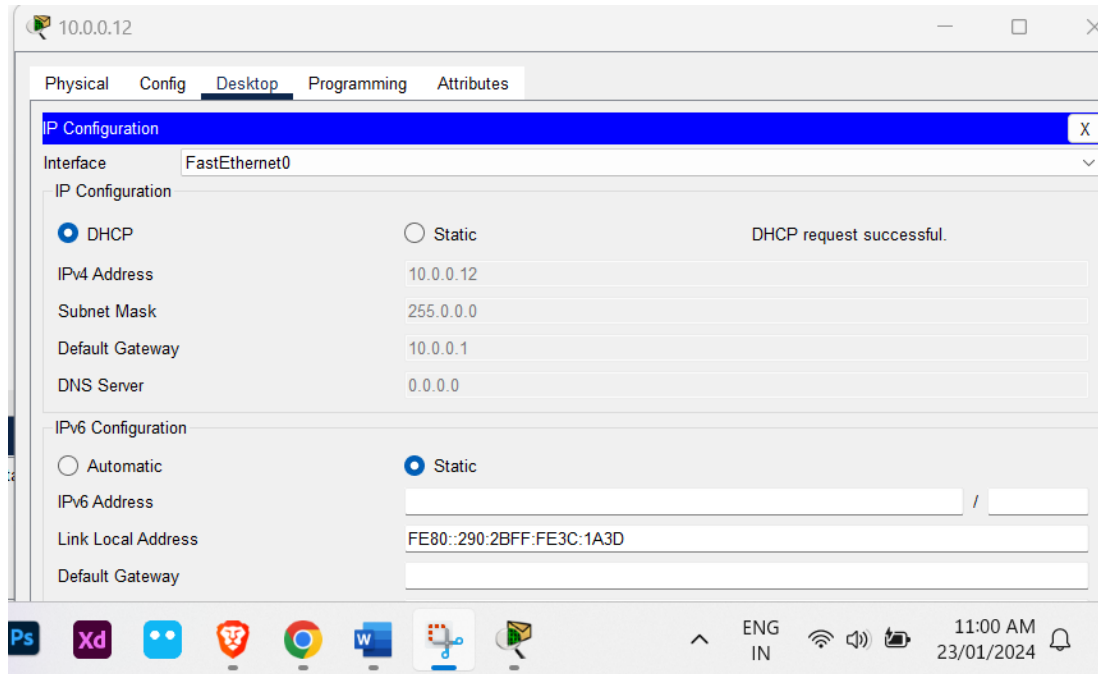
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Step 4: After assigning Ip to the router and server, we setup IP in PC's as:



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Step 5: At last, we ping each nodes to verify the network.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.11

Pinging 10.0.0.11 with 32 bytes of data:

Reply from 10.0.0.11: bytes=32 time<1ms TTL=128
Reply from 10.0.0.11: bytes=32 time=4ms TTL=128
Reply from 10.0.0.11: bytes=32 time<1ms TTL=128
Reply from 10.0.0.11: bytes=32 time=4ms TTL=128

Ping statistics for 10.0.0.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 2ms

C:\>
```

```
C:\>ping 10.0.0.12

Pinging 10.0.0.12 with 32 bytes of data:

Reply from 10.0.0.12: bytes=32 time<1ms TTL=128
Reply from 10.0.0.12: bytes=32 time=6ms TTL=128
Reply from 10.0.0.12: bytes=32 time=6ms TTL=128
Reply from 10.0.0.12: bytes=32 time=1ms TTL=128

Ping statistics for 10.0.0.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 6ms, Average = 3ms

C:\>
```

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```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.13

Pinging 10.0.0.13 with 32 bytes of data:

Reply from 10.0.0.13: bytes=32 time=2ms TTL=128
Reply from 10.0.0.13: bytes=32 time<1ms TTL=128
Reply from 10.0.0.13: bytes=32 time=5ms TTL=128
Reply from 10.0.0.13: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.13:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 5ms, Average = 1ms

C:\>
```

Conclusion:

Hence, we configured DHCP in router and distributed dynamic IP address.



Lab Program Number: ...10...

Date: 2080-10-08

Title: To configure and understand about Routing Information Protocol Using Cisco Packet Tracer.

Theory:

Routing Information Protocol (RIP) is a protocol that routers can use to exchange network topology information. It is characterized as an interior gateway protocol, and is typically used in small to medium-sized networks.

Procedure:

Step 1: First, open the Cisco Packet Tracer desktop and select the devices given below:

S.No	Device	Model Name	Qty.
01	PC	PC	06
02	Switch	PT-Switch	03
03	Router	Router-PT	03

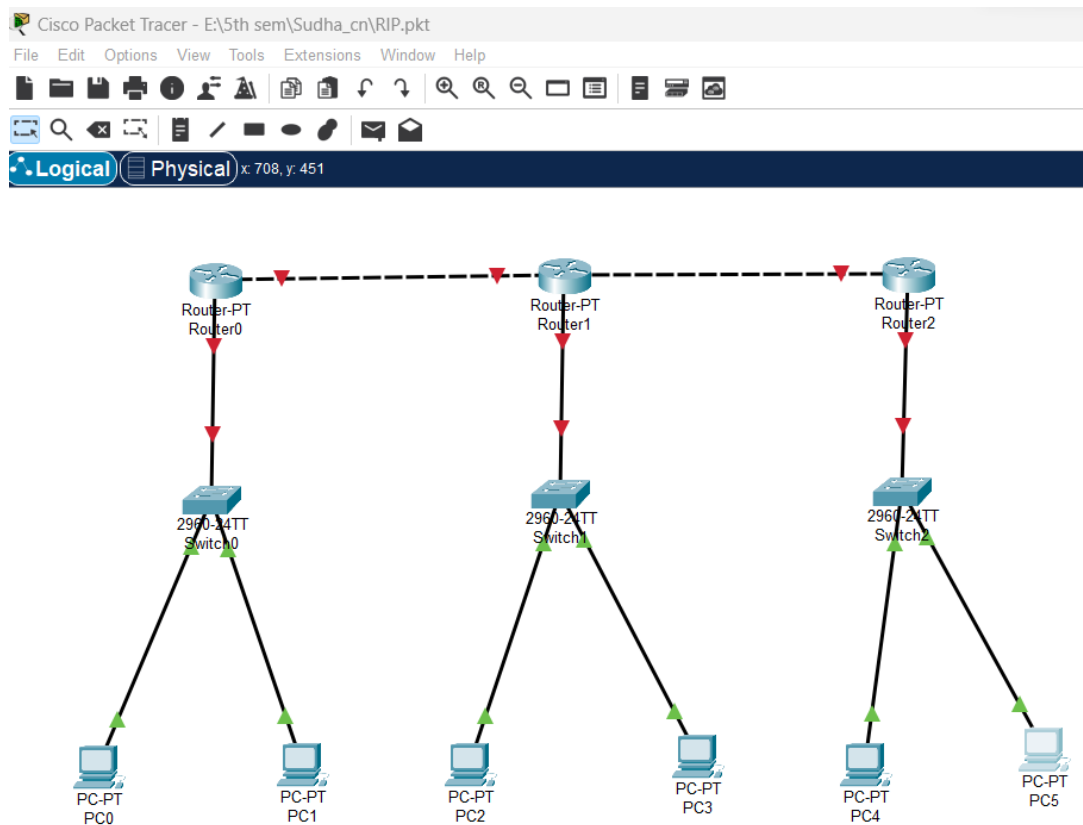
Ip Addressing Table:

S.No	Device	IPv4 Address	Subnet mask	Default Gateway
1	PC0	192.168.10.2	255.255.255.0	192.168.10.1
2	PC1	192.168.10.3	255.255.255.0	192.168.10.1
3	PC2	192.168.20.2	255.255.255.0	192.168.20.1
4	PC3	192.168.20.3	255.255.255.0	192.168.20.1
5	PC4	192.168.30.2	255.255.255.0	192.168.30.1
6	PC5	192.168.30.3	255.255.255.0	192.168.30.1

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- In the “Devices” tab on the left, select “End Devices” and drag a “Generic” Pc or laptop to the workspace.
- Similarly, drag a “Switch” and “Router” to the workspace.
- Click on the “Connections” icon on the left-hand side of the screen.
- Choose the appropriate cable type usually, Ethernet connections use straight-through cables for different devices and cross over for same devices.
- Click on one device’s interface port and then click on the other devices interface to establish a connection.
- Then, create a network topology as shown below the image.

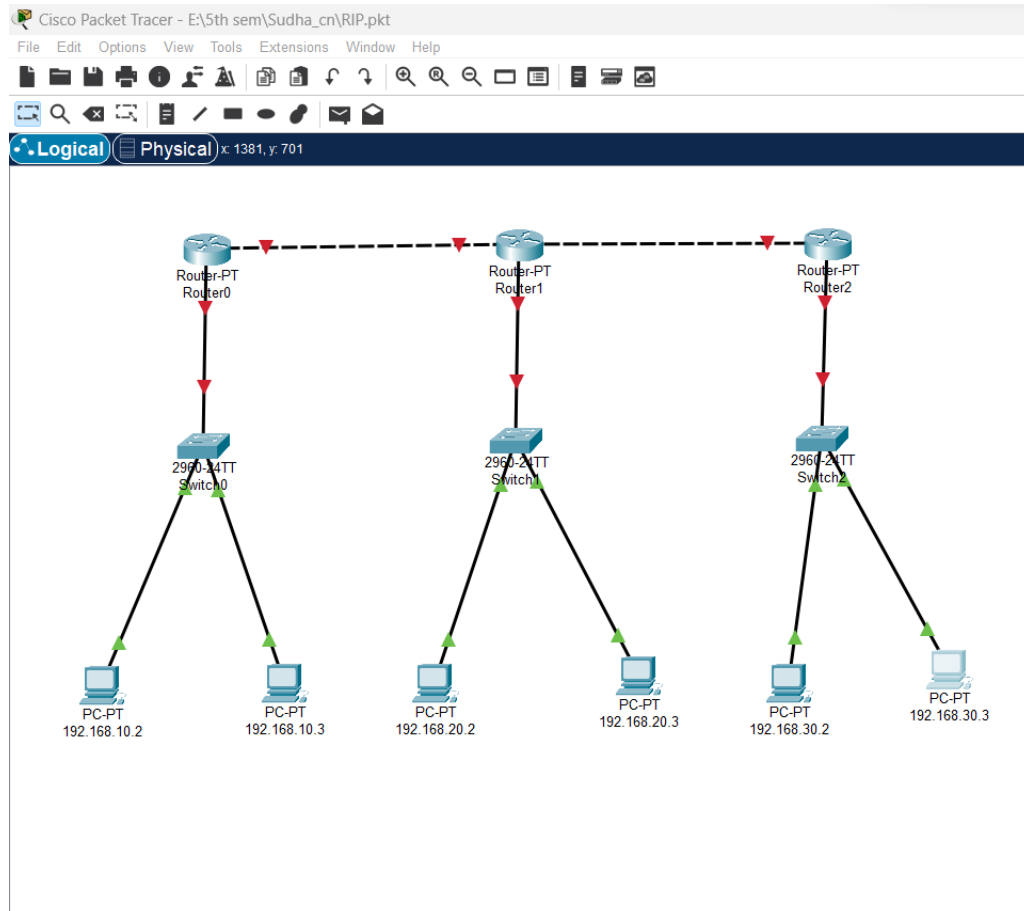


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Step 2: Configure the PCs (hosts) with IPv4 address and Subnet Mask according to the IP addressing table given above.

- To assign an IP address in all PCs.
- Then, go to desktop and then IP configuration and there you will IPv4 configuration.
- Fill IPv4 address and subnet mask.



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- PC0:

192.168.10.2

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.10.2

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

ENG IN 1:06 PM 23/01/2024

- PC1:

192.168.10.3

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.10.3

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

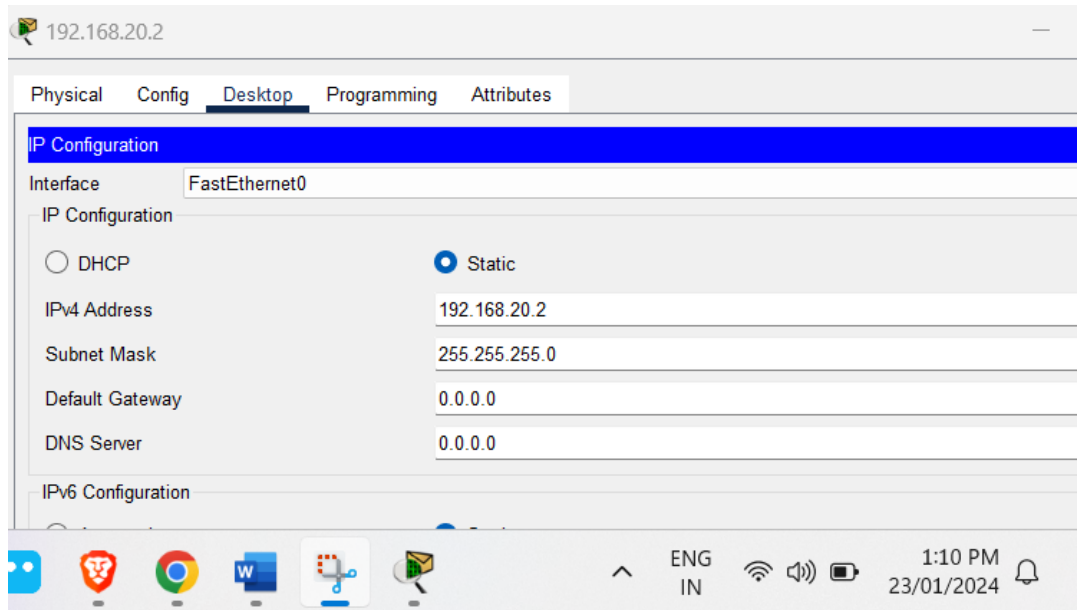
IPv6 Configuration

ENG IN 1:07 PM 23/01/2024

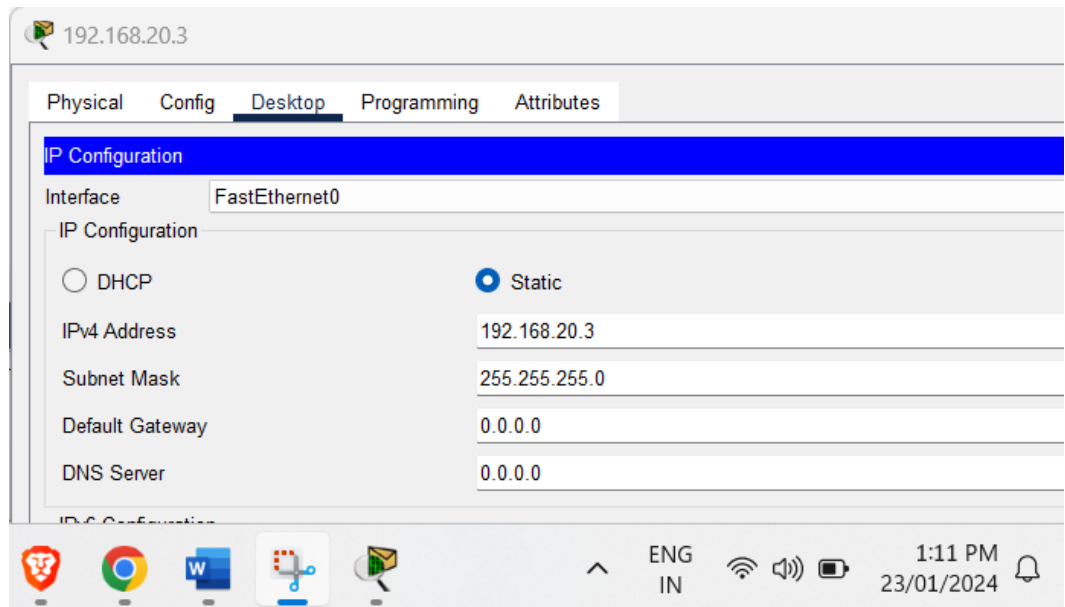
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- PC 2:



- PC3:



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- PC4:

192.168.30.2

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.30.2

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

ENG IN 1:11 PM 23/01/2024

- PC5:

192.168.30.3

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.30.3

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

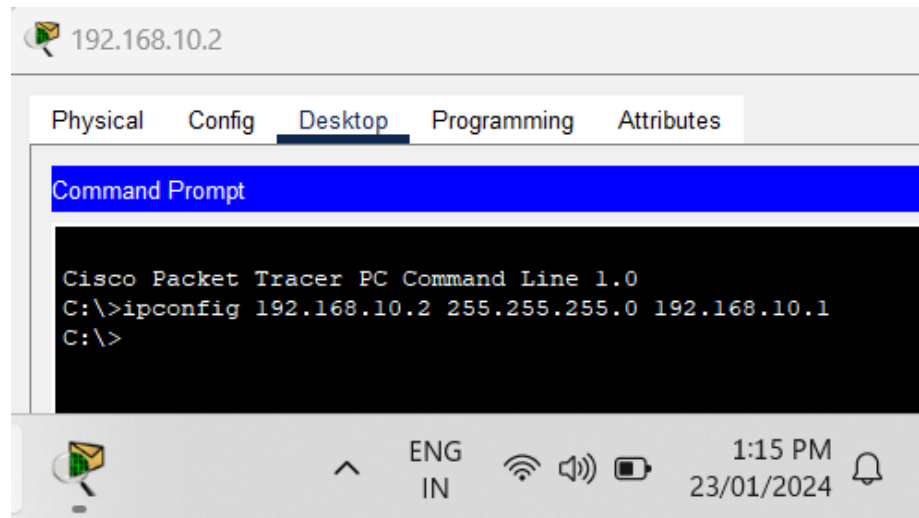
DNS Server 0.0.0.0

ENG IN 1:12 PM 23/01/2024

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- Assigning an IP address using the ipconfig command, or we can also assign an IP address with the help of a command.
- Go to the command terminal of the PC.
- Then, type ipconfig <IPv4 address><subnet mask><default gateway> (if needed)



- Repeat the same procedure with other routers to configure them thoroughly.

Step 3: Configure router with IP address and Subnet mask.

- To assign an IP address in router, click on router.
- Then, go to config and then Interfaces.
- Make sure to turn on the ports.
- Then, configure the IP address in FastEthernet and serial ports according to IP addressing Table.
- Fill IPv4 address and subnet mask for all router.



Router0

Physical Config CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

INTERFACE

- FastEthernet0/0**
- FastEthernet1/0
- Serial2/0
- Serial3/0
- FastEthernet4/0

FastEthernet0/0

Port Status ☒ On

Bandwidth ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☒ Half Duplex ☐ Full Duplex ☒ Auto

MAC Address 0060.5CA3.30D1

IP Configuration

IPv4 Address 192.168.10.1

Subnet Mask 255.255.255.0

Tx Ring Limit 10

Ps Xd [Icons] ENG IN 1:28 PM 23/01/2024

Router1

Physical Config CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

INTERFACE

- FastEthernet0/0**
- FastEthernet1/0
- Serial2/0
- Serial3/0
- FastEthernet4/0
- FastEthernet5/0

FastEthernet0/0

Port Status ☒ On

Bandwidth ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☒ Half Duplex ☐ Full Duplex ☒ Auto

MAC Address 0002.16B9.A6C2

IP Configuration

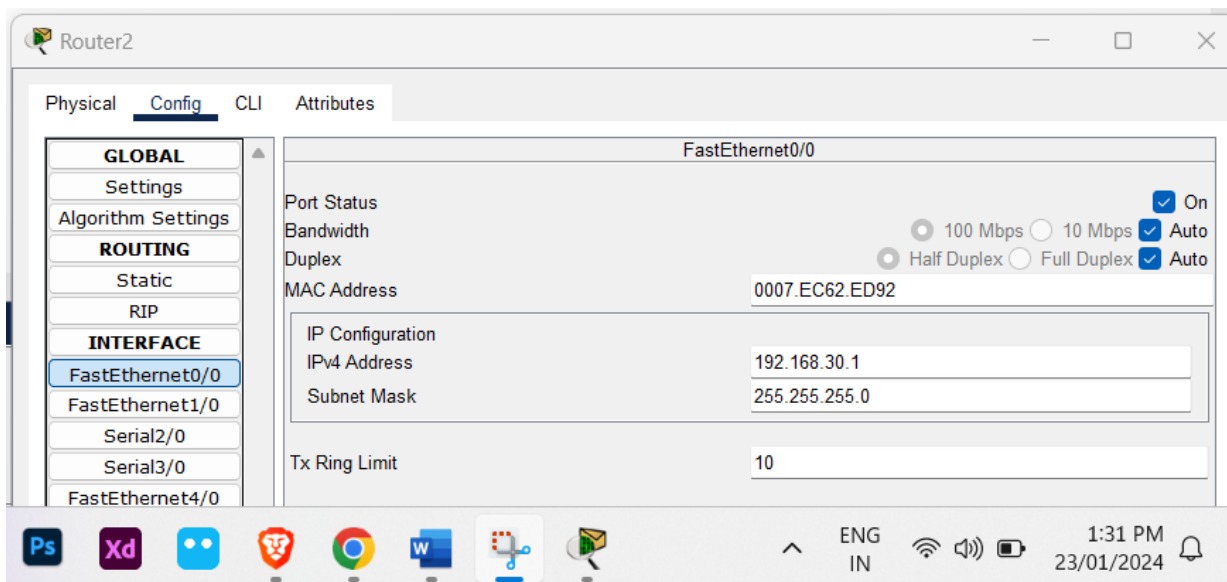
IPv4 Address 192.168.20.1

Subnet Mask 255.255.255.0

Tx Ring Limit 10

Ps Xd [Icons] ENG IN 1:30 PM 23/01/2024

Name: Sudha Pantha



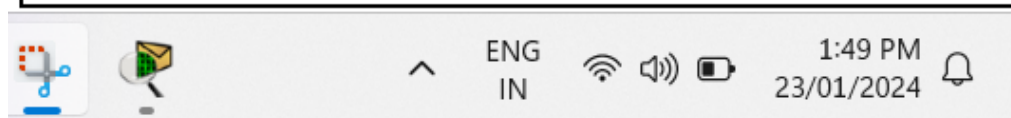
Step 4: After configuring all of the devices we need to assign the routes to the router.

To assign RIP routes to the particular router:

- First, click on router then Go to CLI.
- Then type the commands and IP information given below.
 - CLI command: router rip
 - CLI command: network <network id>

RIP Routes for Router0 are given below:

```
Router>en
Router#conf
Configuring from terminal, memory, or network [terminal]? y
?Must be "terminal", "memory" or "network"
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#network 192.168.10.1
Router(config-router)#network 192.168.30.1
Router(config-router)#do write
Building configuration...
[OK]
Router(config-router)#
```





RIP Routes for Router1 are given below:

```
Router>enable
Router#config
Configuring from terminal, memory, or network [terminal]? y
?Must be "terminal", "memory" or "network"
Router#config t
% Unknown command or computer name, or unable to find computer address

Router#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#network 192.168.20.0
Router(config-router)#10.0.0.0
^
% Invalid input detected at '^' marker.

Router(config-router)#network 10.0.0.0
Router(config-router)#network 11.0.0.0
Router(config-router)#
```

RIP Routes for Router2 are given below:

```
Router2

Press RETURN to get started!

Router>enable
Router#config
Configuring from terminal, memory, or network [terminal]? y
?Must be "terminal", "memory" or "network"
Router#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#network 192.168.30.0
Router(config-router)#network 11.0.0.0
Router(config-router)#
```

Copy Paste

Step 5: Verifying the network by pinging the IP address of any PC.



- PC0:

```
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time=8ms TTL=128
Reply from 192.168.10.2: bytes=32 time=2ms TTL=128
Reply from 192.168.10.2: bytes=32 time=4ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 8ms, Average = 3ms
```

- PC1:

```
C:\>ipconfig 192.168.10.3 255.255.255.0 192.168.10.1
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time=2ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time=6ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 6ms, Average = 2ms

C:\>
```

- PC2:

```
192.168.20.2

Physical Config Desktop Programming Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time=2ms TTL=128
Reply from 192.168.20.2: bytes=32 time=7ms TTL=128
Reply from 192.168.20.2: bytes=32 time=8ms TTL=128
Reply from 192.168.20.2: bytes=32 time=5ms TTL=128

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 8ms, Average = 5ms

C:\>
```



- PC3:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.3

Pinging 192.168.20.3 with 32 bytes of data:

Reply from 192.168.20.3: bytes=32 time=4ms TTL=128
Reply from 192.168.20.3: bytes=32 time=5ms TTL=128
Reply from 192.168.20.3: bytes=32 time=4ms TTL=128
Reply from 192.168.20.3: bytes=32 time=14ms TTL=128

Ping statistics for 192.168.20.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 14ms, Average = 6ms

C:\>
```

- PC4:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.2

Pinging 192.168.30.2 with 32 bytes of data:

Reply from 192.168.30.2: bytes=32 time=9ms TTL=128
Reply from 192.168.30.2: bytes=32 time<1ms TTL=128
Reply from 192.168.30.2: bytes=32 time=5ms TTL=128
Reply from 192.168.30.2: bytes=32 time=5ms TTL=128

Ping statistics for 192.168.30.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 9ms, Average = 4ms

C:\>
```

- PC5:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.3

Pinging 192.168.30.3 with 32 bytes of data:

Reply from 192.168.30.3: bytes=32 time=3ms TTL=128
Reply from 192.168.30.3: bytes=32 time=4ms TTL=128
Reply from 192.168.30.3: bytes=32 time=4ms TTL=128
Reply from 192.168.30.3: bytes=32 time=5ms TTL=128

Ping statistics for 192.168.30.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 5ms, Average = 4ms

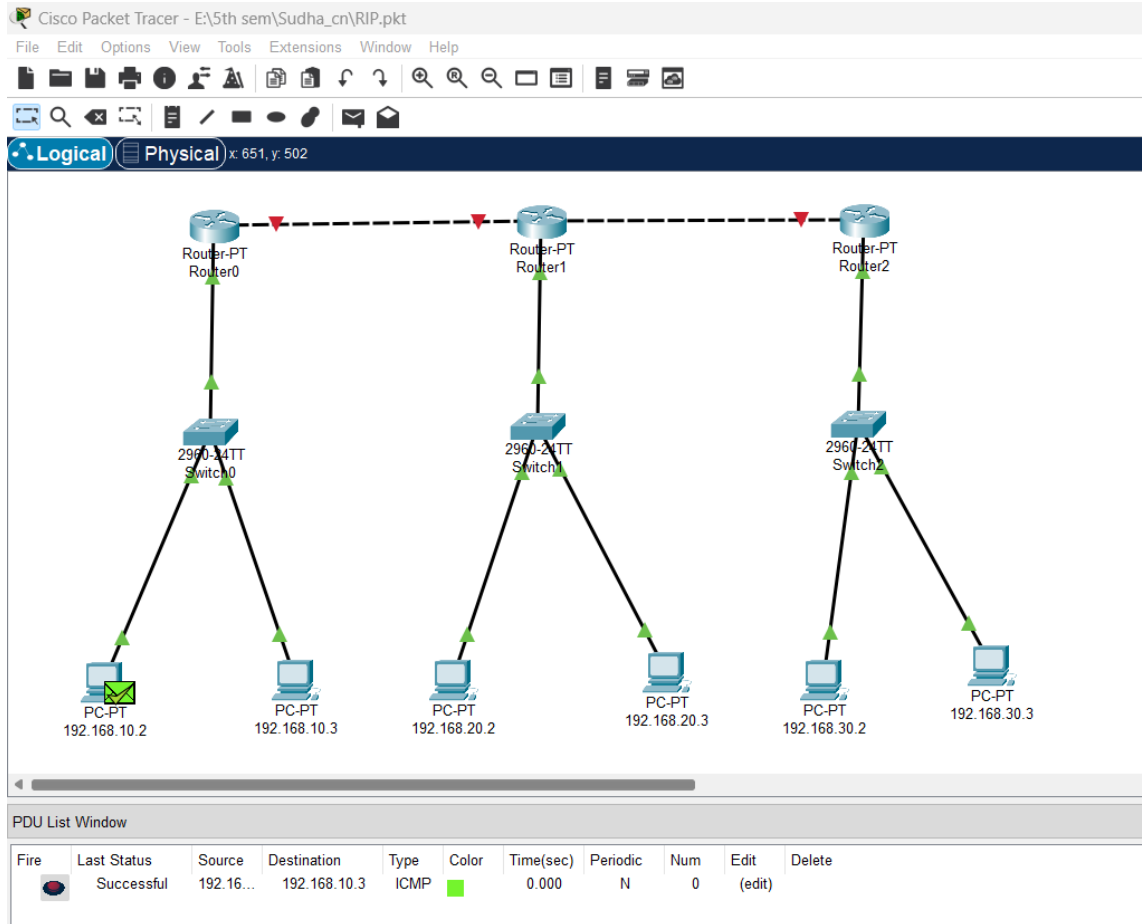
C:\>
```

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Step 6: Simulation

- A simulation of the experiment is given below we are sending PDU from PC0 to PC1:



- Verifying the network by pinging the IP address of any PC.

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```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time=3ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time=3ms TTL=128
Reply from 192.168.10.2: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 2ms

C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Conclusion:

Hence, we configured RIP protocol successfully verified the connection.

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```
Administrator: Command Prompt

C:\Windows\System32>ipconfig/all

Windows IP Configuration

    Host Name . . . . . : DESKTOP-TNG5SP0
    Primary Dns Suffix . . . . . :
    Node Type . . . . . : Mixed
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No

Wireless LAN adapter Wi-Fi:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :
    Description . . . . . : Intel(R) Wi-Fi 6 AX201 160MHz
    Physical Address. . . . . : 20-C1-9B-95-52-81
    DHCP Enabled. . . . . : No
    Autoconfiguration Enabled . . . . : Yes
```

Name: Sudha Pantha



Lab Program Number: ...11...

Date: 2080-10-08

Title: To configure and understand about VLAN using cisco packet tracer.

Theory:

VLANs allow network administrators to automatically limit access to a specified group of users by dividing workstations into different isolated LAN segments. When users move their workstations, administrators don't need to reconfigure the network or change VLAN groups.

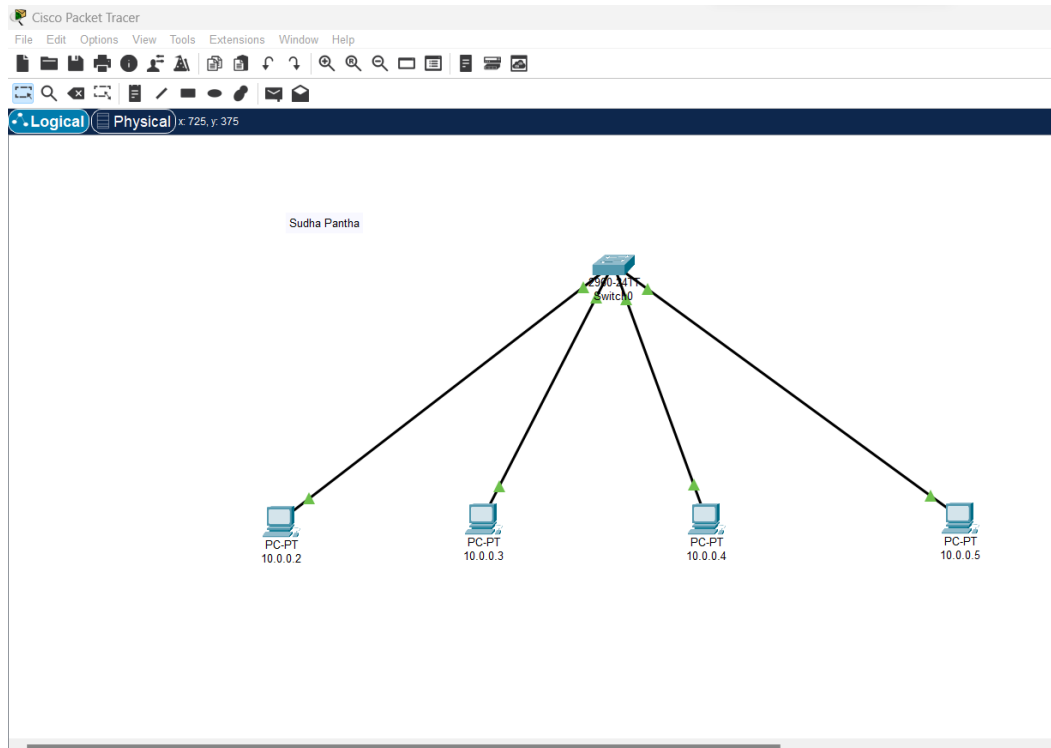
Procedure:

Step 1: Open Cisco Packet Tracer desktop and select all the devices given below:

S.No	Device	Model Name	Qty.
01	PC	PC	04
02	Switch	PT-Switch	01

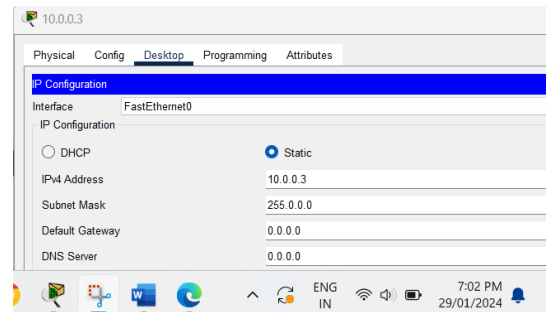
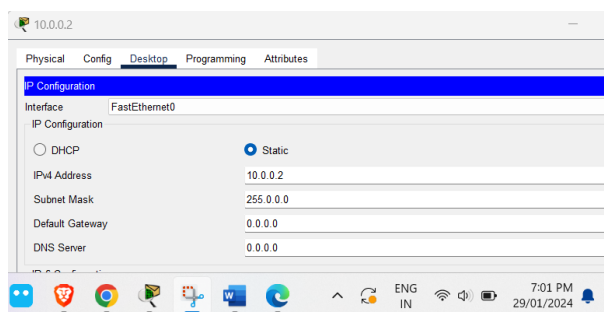
- In the “Devices” tab on the left, select “End Devices” and drag a “Generic” Pc or laptop to the workspace.
- Similarly, drag a “Switch” to the workspace.
- Click on the “Connections” icon on the left-hand side of the screen.
- Choose the appropriate cable type usually, Ethernet connections use straight-through cables.
- Click on one device’s interface port and then click on the other devices interface to establish a connection.
- Then, create a network topology as shown below the image.

Name: Sudha Pantha

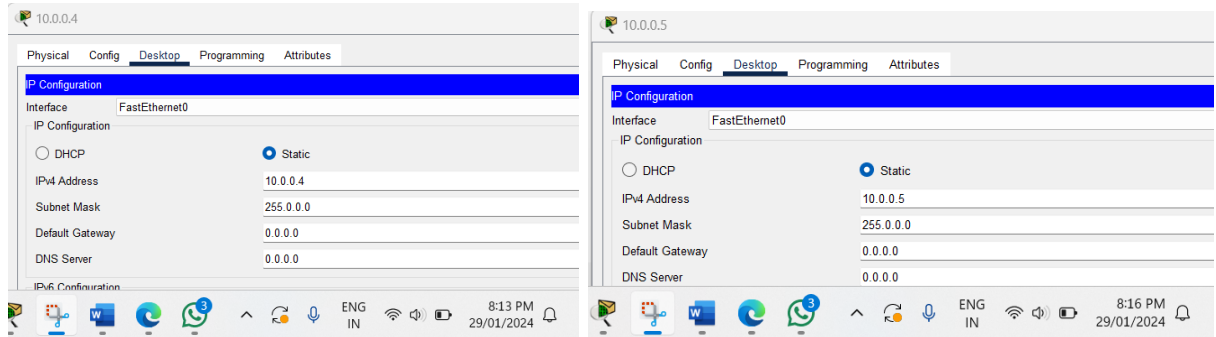


Step 2: Configure the PCs (hosts) with IPv4 address and Subnet Mask according to the IP addressing table given above.

- To assign an IP address in all PCs.
- Then, go to desktop and then IP configuration and there you will IPv4 configuration.
- Fill IPv4 address and subnet mask.

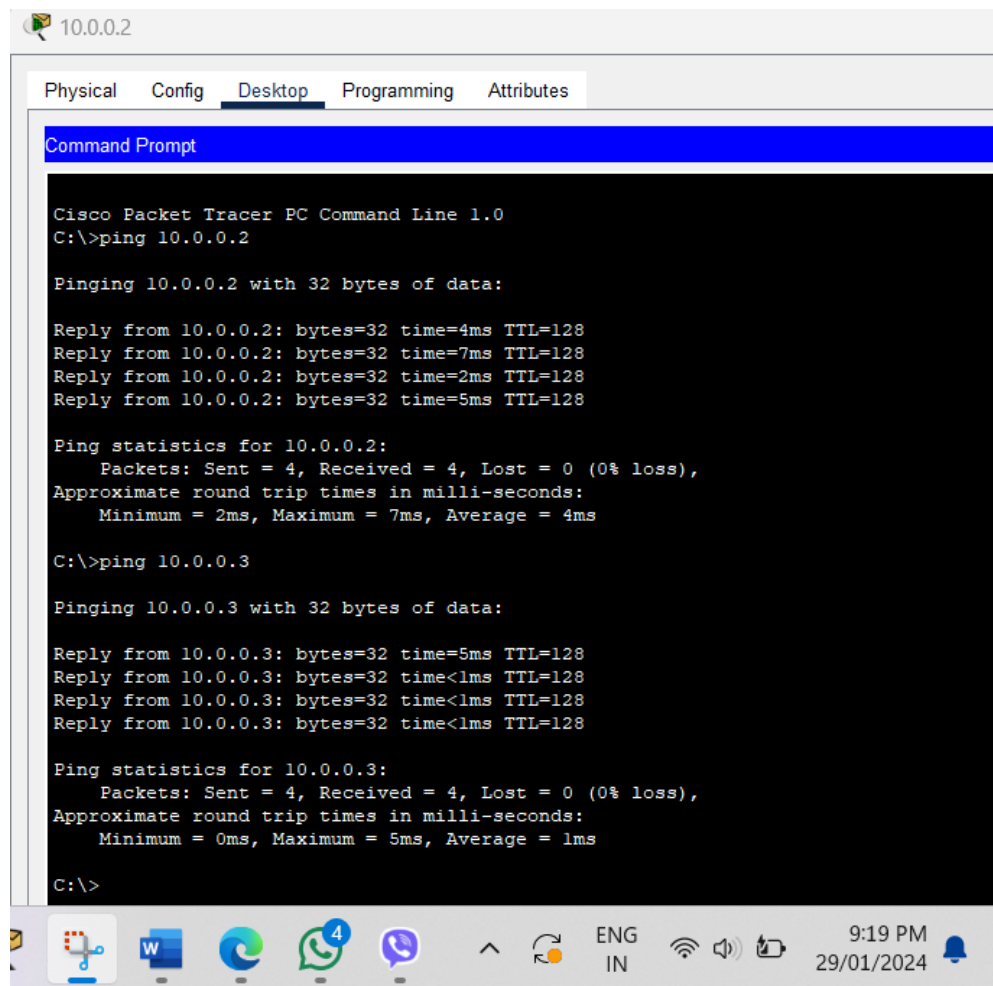


Name: Sudha Pantha



Step 3: Verifying the network by pinging the IP address of any PC.

In PC0 pinging the Ip of PC1 which is reliable.



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```
10.0.0.3
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.3

Pinging 10.0.0.3 with 32 bytes of data:

Reply from 10.0.0.3: bytes=32 time=5ms TTL=128
Reply from 10.0.0.3: bytes=32 time=5ms TTL=128
Reply from 10.0.0.3: bytes=32 time=8ms TTL=128
Reply from 10.0.0.3: bytes=32 time=6ms TTL=128

Ping statistics for 10.0.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 8ms, Average = 6ms

C:\>
```

```
10.0.1.5
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.1.5

Pinging 10.0.1.5 with 32 bytes of data:

Reply from 10.0.1.5: bytes=32 time=2ms TTL=128
Reply from 10.0.1.5: bytes=32 time=8ms TTL=128
Reply from 10.0.1.5: bytes=32 time=6ms TTL=128
Reply from 10.0.1.5: bytes=32 time=7ms TTL=128

Ping statistics for 10.0.1.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 8ms, Average = 5ms

C:\>
```

```
C:\>ping 10.0.1.4

Pinging 10.0.1.4 with 32 bytes of data:

Reply from 10.0.1.4: bytes=32 time=1ms TTL=128
Reply from 10.0.1.4: bytes=32 time=8ms TTL=128
Reply from 10.0.1.4: bytes=32 time=7ms TTL=128
Reply from 10.0.1.4: bytes=32 time=2ms TTL=128

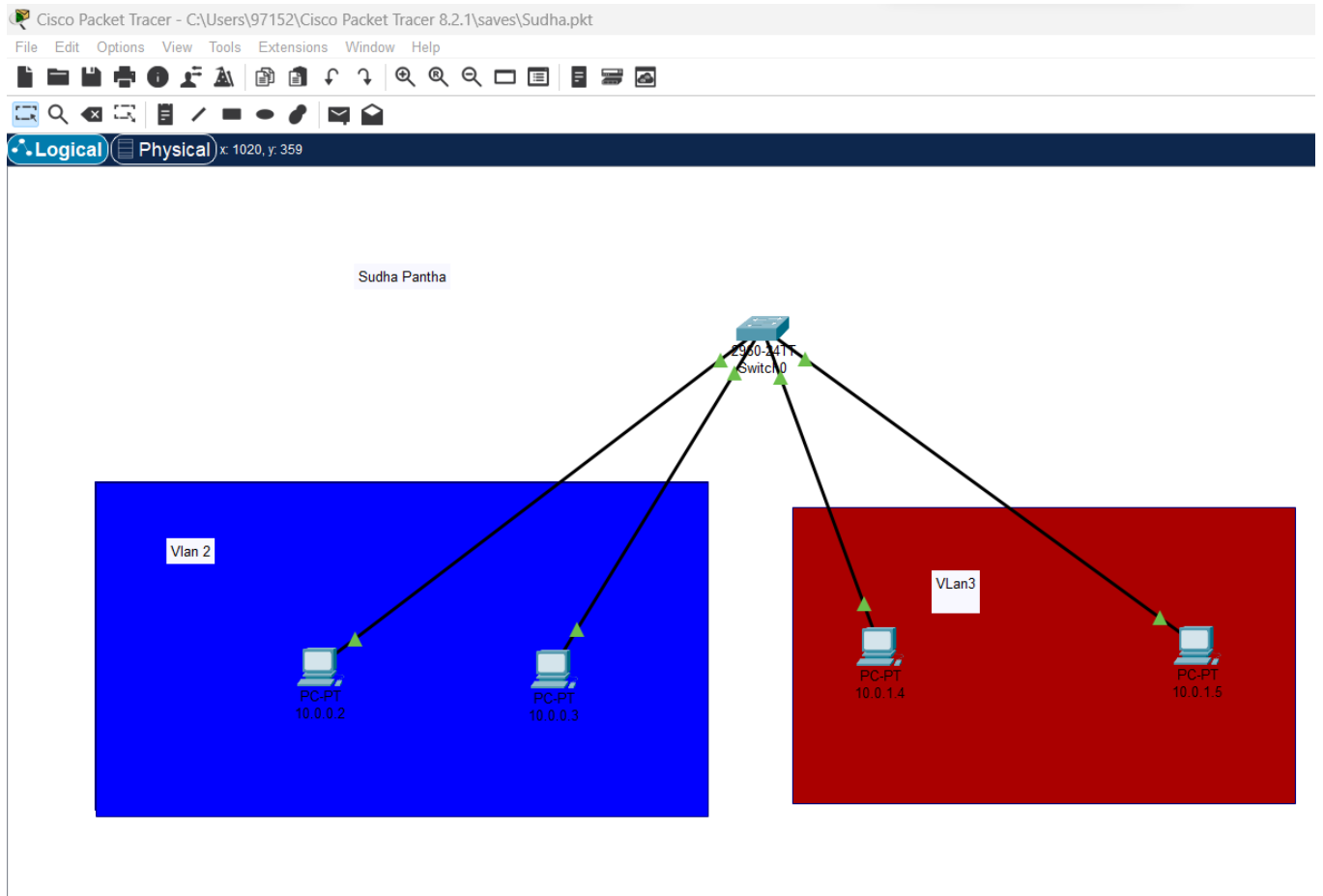
Ping statistics for 10.0.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 8ms, Average = 4ms

C:\>
```

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Step 4: Creating Vlan 2 and Vlan 3



Step 5: Configuring the setting as:

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Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch>enable
Switch#
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#enable
% Incomplete command.
Switch(config)#vlan2
      ^
% Invalid input detected at '^' marker.

Switch(config)#vlan 2
Switch(config-vlan)#name vlan2
Switch(config-vlan)#exit
Switch(config)#vlan 3
Switch(config-vlan)#name vlan3
Switch(config-vlan)#exit
Switch(config)#interface gigabitEthernet 0/1
Switch(config-if)#switchport mode trunk
      ^
% Invalid input detected at '^' marker.

Switch(config-if)#switchport mode trunk
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#switchport access vlan 2
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/2
      ^
% Invalid input detected at '^' marker.

Switch(config)#interface fastEthernet 0/2
      ^
% Invalid input detected at '^' marker.

Switch(config)#interface fastEthernet 0/2
Switch(config-if)#switchport access vlan 3
Switch(config-if)#exit
```

Copy Paste

☐ Top

90%

9:44 PM 29/01/2024

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```
Switch0
Physical Config CLI Attributes
IOS Command Line Interface

Press RETURN to get started.

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 2
Switch(config-vlan)#name vlan2
Switch(config-vlan)#exit
Switch(config)#vlan 3
Switch(config-vlan)#name vlan3
Switch(config-vlan)#exit
Switch(config)#interface gigabitEthernet 0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#interface fastEthernet0/1
Switch(config-if)#switchport access vlan1
Switch(config-if)#switchport access vlan 2
Switch(config-if)#exit
Switch(config)#interface fastEthernet0/2
Switch(config-if)#switchport access vlan 3
Switch(config-if)#exit
Switch(config)#do write
Building configuration...
[OK]
Switch(config)#
```

Step 6: Finally, we ping computers and verify network in vlan

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10.0.0.3

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.3

Pinging 10.0.0.3 with 32 bytes of data:

Reply from 10.0.0.3: bytes=32 time=8ms TTL=128
Reply from 10.0.0.3: bytes=32 time=5ms TTL=128
Reply from 10.0.0.3: bytes=32 time=8ms TTL=128
Reply from 10.0.0.3: bytes=32 time=6ms TTL=128

Ping statistics for 10.0.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 8ms, Average = 6ms

C:\> ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Taskbar icons: File Explorer, Network, Word, Edge, WhatsApp (3), Telegram, Up arrow, Refresh, ENG IN, Wi-Fi, Speaker, Battery, 10:25 PM 29/01/2024, Notification bell.

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The screenshot shows a Cisco Packet Tracer PC Command Line window for a device named 10.0.0.5. The window has tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying a Command Prompt. The Command Prompt shows the following output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.5

Pinging 10.0.0.5 with 32 bytes of data:

Reply from 10.0.0.5: bytes=32 time<1ms TTL=128
Reply from 10.0.0.5: bytes=32 time=1ms TTL=128
Reply from 10.0.0.5: bytes=32 time=3ms TTL=128
Reply from 10.0.0.5: bytes=32 time=4ms TTL=128

Ping statistics for 10.0.0.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 2ms

C:\>ping 10.0.0.4

Pinging 10.0.0.4 with 32 bytes of data:

Reply from 10.0.0.4: bytes=32 time=11ms TTL=128
Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Reply from 10.0.0.4: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 2ms

C:\>ping 10.0.0.3

Pinging 10.0.0.3 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
```

The Windows taskbar at the bottom shows the time as 10:27 PM on 29/01/2024, along with various system icons and application shortcuts.

Conclusion:

Hence, we configured vlan and confirmed the connection.

Name: Sudha Pantha



Lab Program Number: ...12...

Date: 2080-10-08

Title: To configure and understand about router on a stick using cisco packet tracer.

Theory:

ROUTER-ON-A-STICK, also known as a “one-armed router” is a method for running multiple VLANs over a single connection in order to provide inter-VLAN routing. Essentially the router connects to a core switch with a single interface and acts as the relay point between networks.

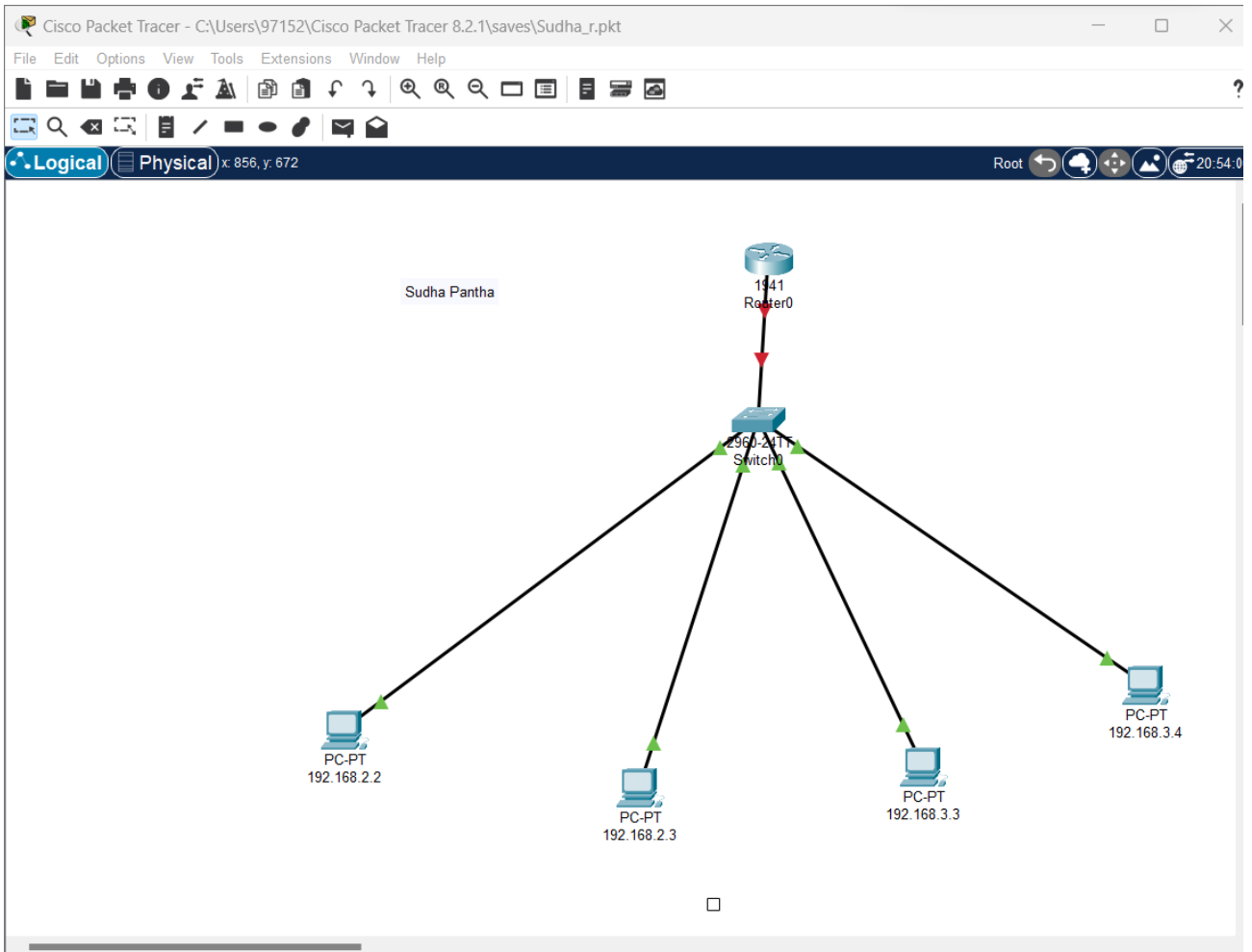
Procedure:

Step 1: Open Cisco Packet Tracer desktop and select all the devices given below:

S.No	Device	Model Name	Qty.
01	PC	PC	04
02	Switch	PT-Switch	01
03	Router	Router-PT	01

- In the “Devices” tab on the left, select “End Devices” and drag a “Generic” Pc or laptop to the workspace.
- Similarly, drag a “Switch” and “Router” to the workspace.
- Click on the “Connections” icon on the left-hand side of the screen.
- Choose the appropriate cable type usually, Ethernet connections use straight-through cables.
- Click on one device’s interface port and then click on the other devices interface to establish a connection.
- Then, create a network topology as shown below the image.

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Step 2: Configure the PCs (hosts) with IPv4 address and Subnet Mask according to the IP addressing table given above.

- To assign an IP address in all PCs.
- Then, go to desktop and then IP configuration and there you will IPv4 configuration.
- Fill IPv4 address and subnet mask.

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192.168.2.2

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.2.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

Chrome Mail W E S ENG IN 2:24 PM 31/01/2024

192.168.2.3

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.2.3

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

Chrome Mail W E S ENG IN 2:24 PM 31/01/2024

Name: Sudha Pantha



192.168.3.3

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.3.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.20.1

DNS Server 0.0.0.0

Taskbar: File Explorer, Word, Edge, Remote Desktop, Network, ENG IN, 2:25 PM 31/01/2024

192.168.3.4

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.3.4

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

Taskbar: File Explorer, Word, Edge, Remote Desktop, Network, ENG IN, 2:29 PM 31/01/2024

Name: Sudha Pantha



Step 3: Verifying the network by pinging the IP address of any PC.

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time=7ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time=4ms TTL=128
Reply from 192.168.2.2: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 3ms

C:\>
```



```
C:\>
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time=2ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time=10ms TTL=128
Reply from 192.168.2.3: bytes=32 time=6ms TTL=128

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 4ms
```



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```
C:\>ping 192.168.3.3

Pinging 192.168.3.3 with 32 bytes of data:

Reply from 192.168.3.3: bytes=32 time<1ms TTL=128
Reply from 192.168.3.3: bytes=32 time<1ms TTL=128
Reply from 192.168.3.3: bytes=32 time=8ms TTL=128
Reply from 192.168.3.3: bytes=32 time=3ms TTL=128

Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
```



```
C:\>ping 192.168.3.4

Pinging 192.168.3.4 with 32 bytes of data:

Reply from 192.168.3.4: bytes=32 time=2ms TTL=128
Reply from 192.168.3.4: bytes=32 time=5ms TTL=128
Reply from 192.168.3.4: bytes=32 time=3ms TTL=128
Reply from 192.168.3.4: bytes=32 time<1ms TTL=128

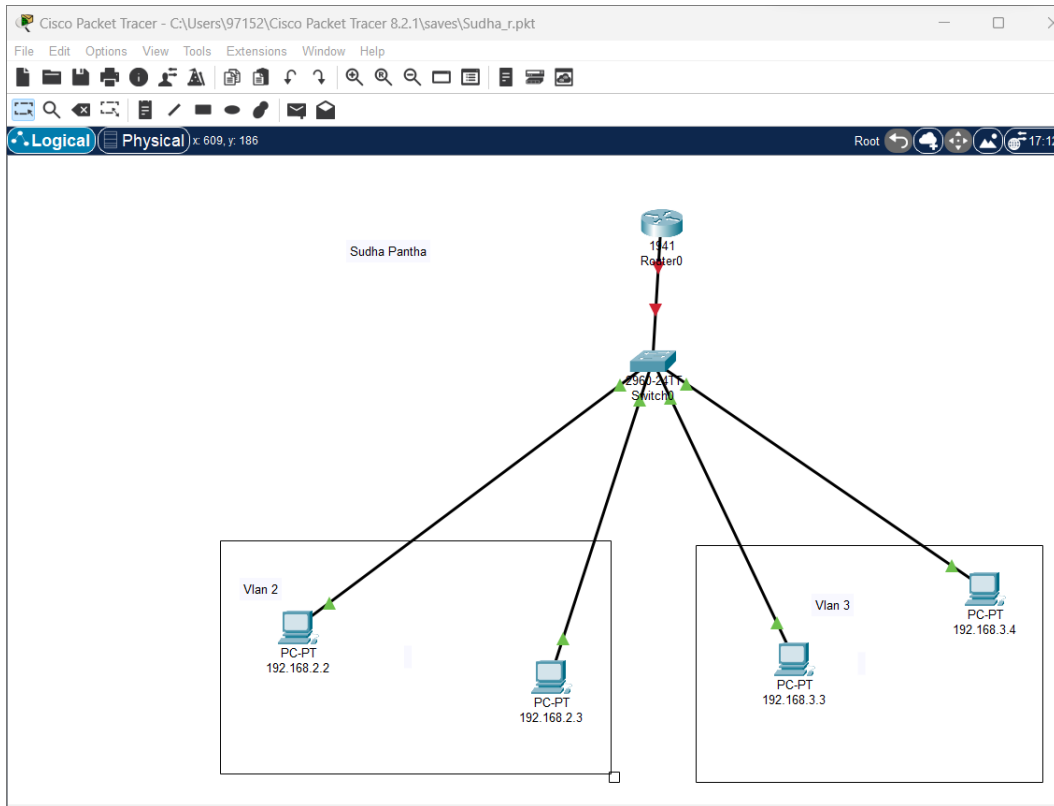
Ping statistics for 192.168.3.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 5ms, Average = 2ms

C:\>
```



Step 4: Then, we configure a topology for our Router-on- stick, as shown below:

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- Then, we setup a Vlan network

```
Switch0
Physical Config CLI Attributes
IOS Command Line Interface

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 3
Switch(config-vlan)#interface fastethernet 0/1
Switch(config-if)#switchport access vlan 2
Switch(config-if)#interface fastethernet 0/2
Switch(config-if)#switchport access vlan 2
Switch(config-if)#interface fastethernet 0/3
Switch(config-if)#switchport access vlan 3
Switch(config-if)#interface fastethernet 0/4
Switch(config-if)#switchport access vlan 3
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#write
Building configuration...
[OK]
Switch#
```

- After setting up vlan, we setup a trunk for our vlan traffic as:

Name: Sudha Pantha



```
Switch#write
Building configuration...
[OK]
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface gigabitEthernet 0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#write
Building configuration...
[OK]
Switch#
```

[Copy](#) [Paste](#)

[Top](#)

- After the trunk, we set up the router

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0.10
Router(config-subif)#encapsulation dot1Q 10
^
% Invalid input detected at '^' marker.

Router(config-subif)#encapsulation dot1Q 10
^
% Invalid input detected at '^' marker.

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#interface gigabitEthernet 0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
% 192.168.10.0 overlaps with GigabitEthernet0/0.10
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#interface gigabitEthernet 0/0
Router(config-if)#no shutdown
Router(config-if)#
```

Conclusion:

Hence, we configured Router on a stick and verified.



Lab Program Number: ...13...

Date: 2080-10-08

Title: To set up web DNS server in cisco packet Tracer.

Theory:

The Domain Name System (DNS) Server is a server that is specifically used for matching website hostnames (like `example.com`) to their corresponding Internet Protocol or IP addresses. The DNS server contains a database of public IP addresses and their corresponding domain names. Every device connected to the internet has a unique IP address that helps to identify it, according to the IPv4 or IPV6 protocols. The same goes for web servers that host websites.

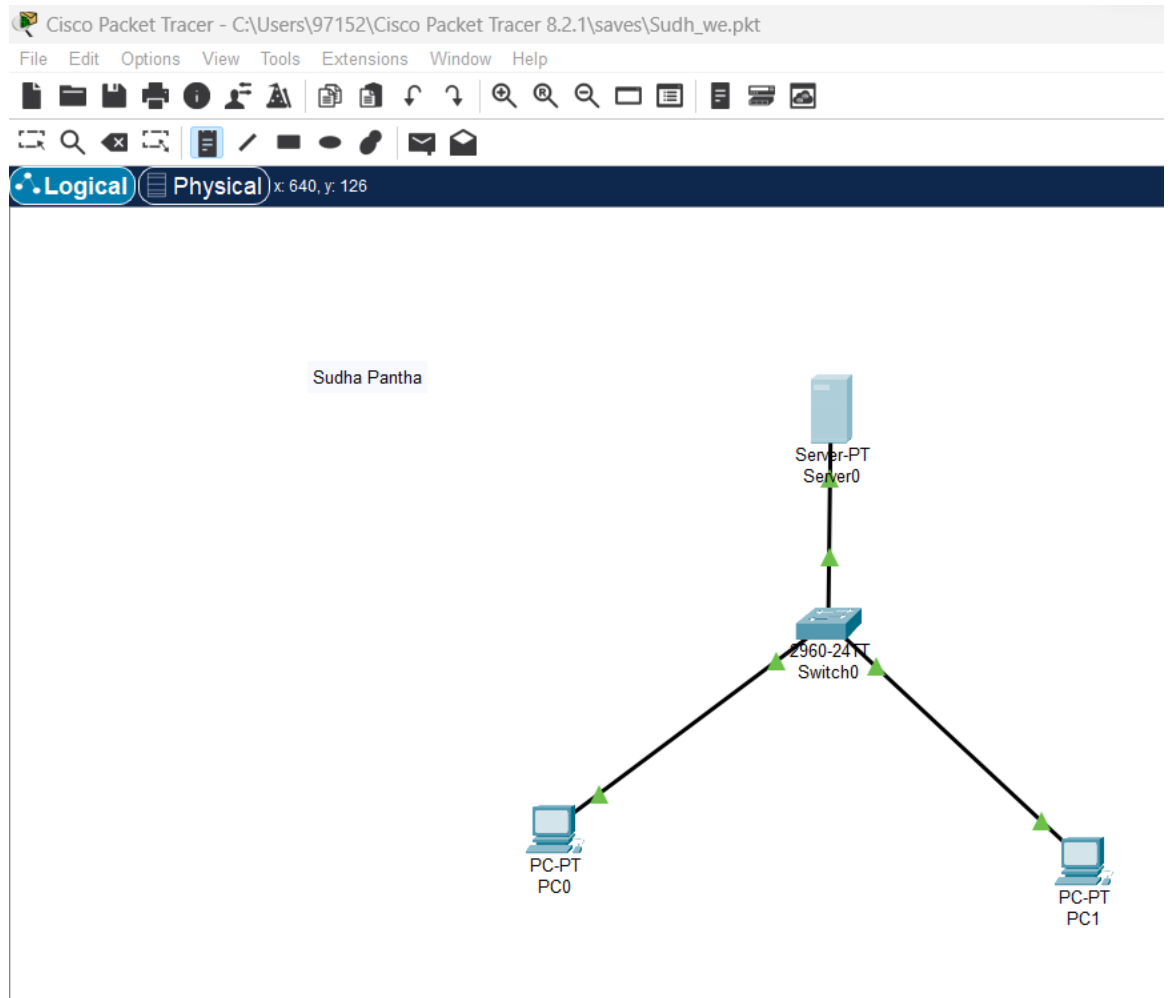
Procedure:

Step 1: First, open the cisco packet tracer desktop and select the devices given below:

S.No	Device	Model Name	Qty.
01	PC	PC	02
02	Switch	PT-Switch	01
03	Server	Server-PT	01

- In the “Devices” tab on the left, select “End Devices” and drag a “Generic” Pc or laptop to the workspace.
- Similarly, drag a “Switch” and “Router” to the workspace.
- Click on the “Connections” icon on the left-hand side of the screen.
- Choose the appropriate cable type usually, Ethernet connections use straight-through cables.
- Click on one device’s interface port and then click on the other devices interface to establish a connection.
- Then, create a network topology as shown below the image.

Name: Sudha Pantha



Step 2: Configure static Ip addresses on the PCs and the server.

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Server0

Physical Config Services **Desktop** Programming Attributes

IP Configuration

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.2.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.2.1

DNS Server 192.168.2.2

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80::202:17FF:FEA6:9AC6

Taskbar: VS Code, Word, WhatsApp (3), Edge, File Explorer, Network icon, ENG US, 8:17 PM, 02/02/2024, Bell icon

PC0

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.2.3

Subnet Mask 255.255.255.0

Default Gateway 198.168.2.1

DNS Server 198.168.2.2

IPv6 Configuration

Taskbar: Chrome, VS Code, Word, WhatsApp (3), Edge, File Explorer, Network icon, ENG US, 8:33 PM, 02/02/2024, Bell icon

Name: Sudha Pantha



PC1

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.2.4

Subnet Mask 255.255.255.0

Default Gateway 192.168.2.1

DNS Server 192.168.2.2

Taskbar: VS Code, Word, WhatsApp (3), Edge, File Explorer, Remote Desktop, ENG US, 8:51 PM 02/02/2024

Step 3: Configure DNS service on the generic server.

- To do this, click on the server, then Click on **Services** tab.
- Click on **DNS server** from the menu.
- First turn **ON** the DNS service, then define **names** of the hosts and their corresponding **IP addresses**.

Server0

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS**
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DNS

DNS Service ☒ On ☐ Off

Resource Records

Name Type A Record

Address

Add Save Remove

No.	Name	Type	Detail
0	dns_server	A Record	192.168.2.2
1	pc0	A Record	192.168.2.2
2	pc1	A Record	192.168.2.3

Taskbar: Ps, Xd, Teams, Chrome, Word, WhatsApp (3), Edge, File Explorer, Remote Desktop, ENG US, 9:09 PM 02/02/2024

Name: Sudha Pantha



Step 4: Verifying the network by pinging the IP address of any PC.

PC0 Command Prompt:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping PC0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time=1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

PC1 Command Prompt:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time=2ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

PC0 Command Prompt:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping PC0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time=1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time=10ms TTL=128
Reply from 192.168.2.3: bytes=32 time=3ms TTL=128
Reply from 192.168.2.3: bytes=32 time=3ms TTL=128
Reply from 192.168.2.3: bytes=32 time=5ms TTL=128

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 10ms, Average = 5ms
```

Conclusion:

Hence, we configured web dns server and verified.

Name: Sudha Pantha



Lab Program Number: ...14...

Date: 2080-10-22

Title: To set up OSPF in cisco packet Tracer.

Theory:

OSPF stands for Open Shortest Path First a link-state protocol and as the name itself justifies that it is used to find the best and the optimal pathway between the starting point and the destination target router using its own shortest path first algorithm. It sends hello packets every 10 seconds. It varies under (LSRP) Link State Routing Protocol. It automatically generates or decides the optimal path and it is used to send router packets with the auto path decision method. It is one of the Interior gateway protocols that aim to move the packet within a large autonomous system. OSPF works on port no. 89 and uses [AD value 110].

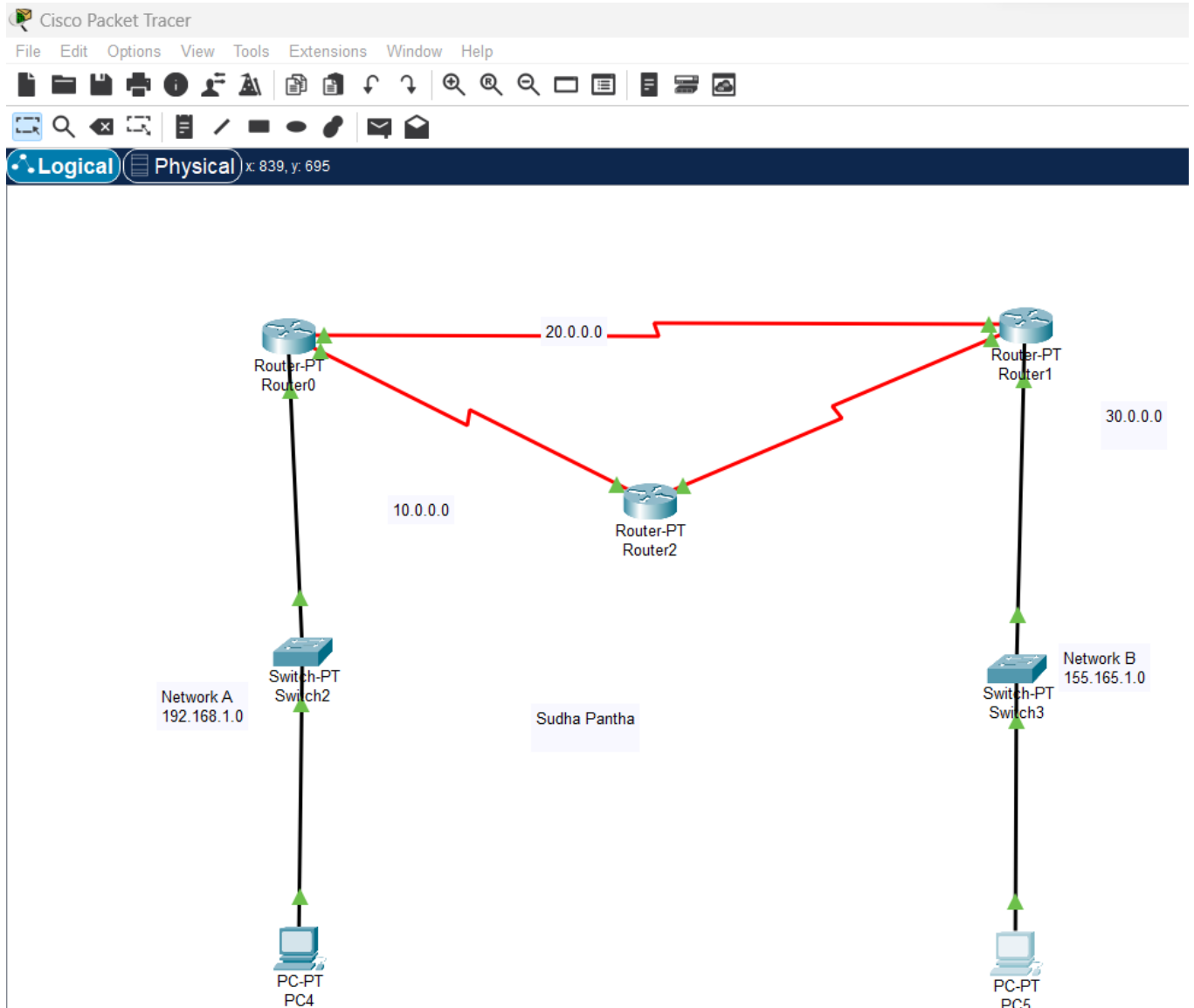
Procedure:

Step 1: First, open the cisco packet tracer desktop and select the devices given below:

S.No	Device	Model Name	Qty.
01	PC	PC	02
02	Switch	PT-Switch	02
03	Router	Router-PT	03

- In the “Devices” tab on the left, select “End Devices” and drag a “Generic” Pc or laptop to the workspace.
- Similarly, drag a “Switch” and “Router” to the workspace.
- Click on the “Connections” icon on the left-hand side of the screen.
- Choose the appropriate cable type usually, Ethernet connections use straight-through cables.
- Click on one device’s interface port and then click on the other devices interface to establish a connection.
- Then, create a network topology as shown below the image.

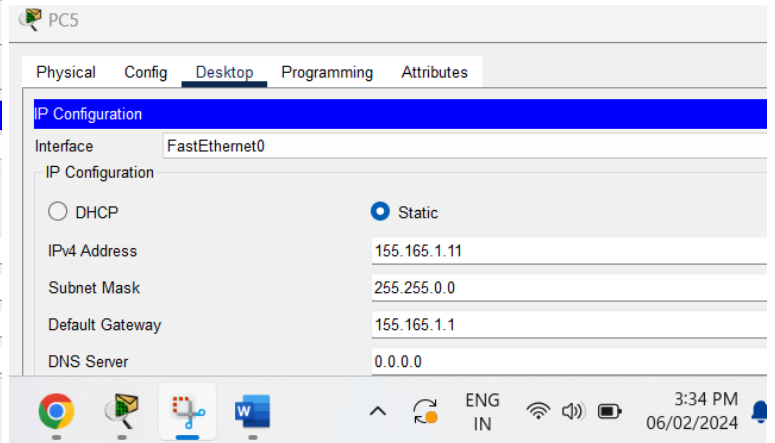
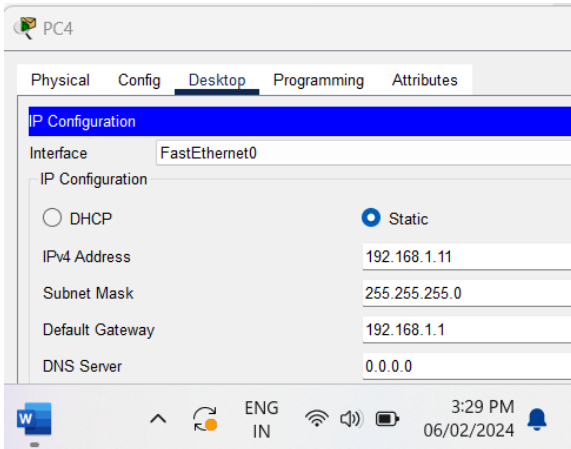
Name: Sudha Pantha



Step 2: Configure the PCs (hosts) with IPv4 address and Subnet Mask according to the IP addressing table given above.

- To assign an IP address in all PCs.
- Then, go to desktop and then IP configuration and there you will IPv4 configuration.
- Fill IPv4 address and subnet mask.

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Step 3: Configure router with IP address and Subnet mask.

- To assign an IP address in router, click on router.
- Then, go to config and then Interfaces.
- Make sure to turn on the ports.
- Then, configure the IP address in FastEthernet and serial ports according to IP addressing Table.
- Fill IPv4 address and subnet mask for all router.



Router0

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings
- ROUTING**
- Static
- RIP
- INTERFACE**
- FastEthernet0/0**
- FastEthernet1/0
- Serial2/0
- Serial3/0
- FastEthernet4/0

FastEthernet0/0

Port Status ☒ On

Bandwidth ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☐ Full Duplex ☒ Auto

MAC Address 000A.F322.1C17

IP Configuration

IPv4 Address 192.168.1.1

Subnet Mask 255.255.255.0

Tx Ring Limit 10

3:38 PM 06/02/2024

Router0

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings
- ROUTING**
- Static
- RIP
- INTERFACE**
- FastEthernet0/0
- FastEthernet1/0
- Serial2/0**
- Serial3/0
- FastEthernet4/0
- FastEthernet5/0

Serial2/0

Port Status ☒ On

Duplex ☐ Full Duplex

Clock Rate Not Set

IP Configuration

IPv4 Address 10.0.0.2

Subnet Mask 255.0.0.0

Tx Ring Limit 10

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Router2

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

INTERFACE

- FastEthernet0/0
- FastEthernet1/0
- Serial2/0
- Serial3/0**
- FastEthernet4/0

Serial3/0

Port Status ☒ On

Duplex ☐ Full Duplex

Clock Rate 64000

IP Configuration

IPv4 Address 30.0.0.1

Subnet Mask 255.0.0.0

Tx Ring Limit 10

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Router1

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

INTERFACE

- FastEthernet0/0
- FastEthernet1/0
- Serial2/0**
- Serial3/0
- FastEthernet4/0
- FastEthernet5/0

Serial2/0

Port Status ☒ On

Duplex ☐ Full Duplex

Clock Rate Not Set

IP Configuration

IPv4 Address 30.0.0.2

Subnet Mask 255.0.0.0

Tx Ring Limit 10

4:02 PM 06/02/2024

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Router2

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

Serial2/0

Port Status

Duplex ☐ Full Duplex

Clock Rate 64000

IP Configuration

IPv4 Address 10.0.0.1

Subnet Mask 255.0.0.0

Tx Ring Limit 10

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Router0

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

Serial3/0

Port Status

Duplex ☐ Full Duplex

Clock Rate 64000

IP Configuration

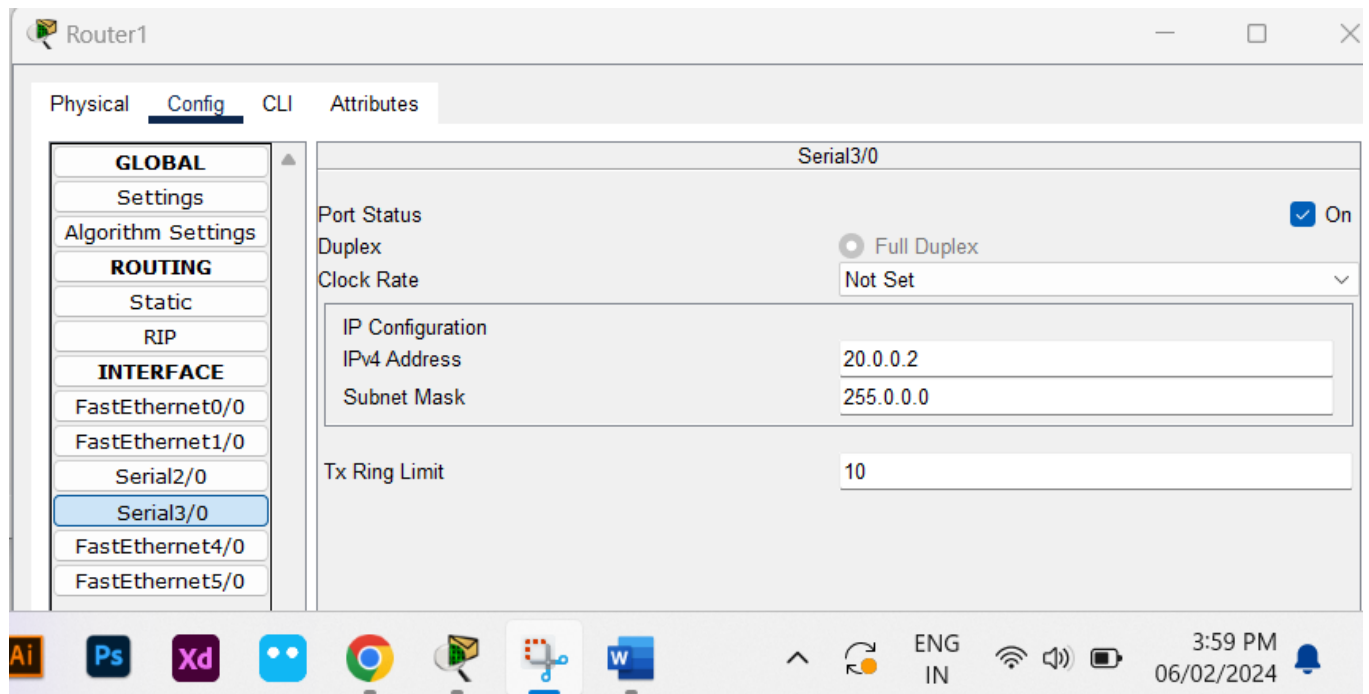
IPv4 Address 20.0.0.1

Subnet Mask 255.0.0.0

Tx Ring Limit 10

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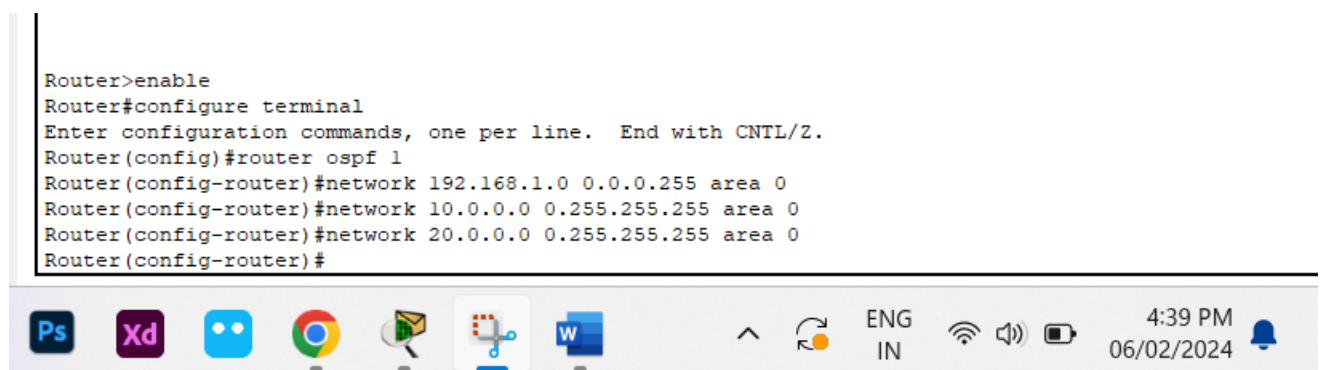


Step 4: After configuring all of the devices we need to assign the routes to the router.

To assign RIP routes to the particular router:

- First, click on router then Go to CLI.
- Then type the commands and IP information given below.
 - CLI command: router rip
 - CLI command: network <network id>

In Router 0:



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Router 2:

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface Serial2/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial2/0
Router(config-if)#ext
^
% Invalid input detected at '^' marker.

Router(config-if)#exit
Router(config)#router ospf 1
Router(config-router)#network 10.0.0.0 0.255.255.255 area 0
Router(config-router)#net
01:24:34: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on Serial2/0 from LOADING to FULL, Loading
Done
work
% Incomplete command.
Router(config-router)#network 30.0.0.0 0.255.255.255 area 0
Router(config-router)#exit
Router(config)#
```

Copy

Paste



In Router 1:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 20.0.0.0 0.255.255.255 area 0
Router(config-router)#network 30.0.0.0
01:30:17: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on Serial3/0 from LOADING to FULL, Load
Done

% Incomplete command.
Router(config-router)#network 30.0.0.0 0.255.255.255 area 0
Router(config-router)#
01:31:21: %OSPF-5-ADJCHG: Process 1, Nbr 30.0.0.1 on Serial2/0 from LOADING to FULL, Loading

Router(config-router)#network 155.165.1.0 0.0.255.255 area 0
Router(config-router)#exit
Router(config)#
```

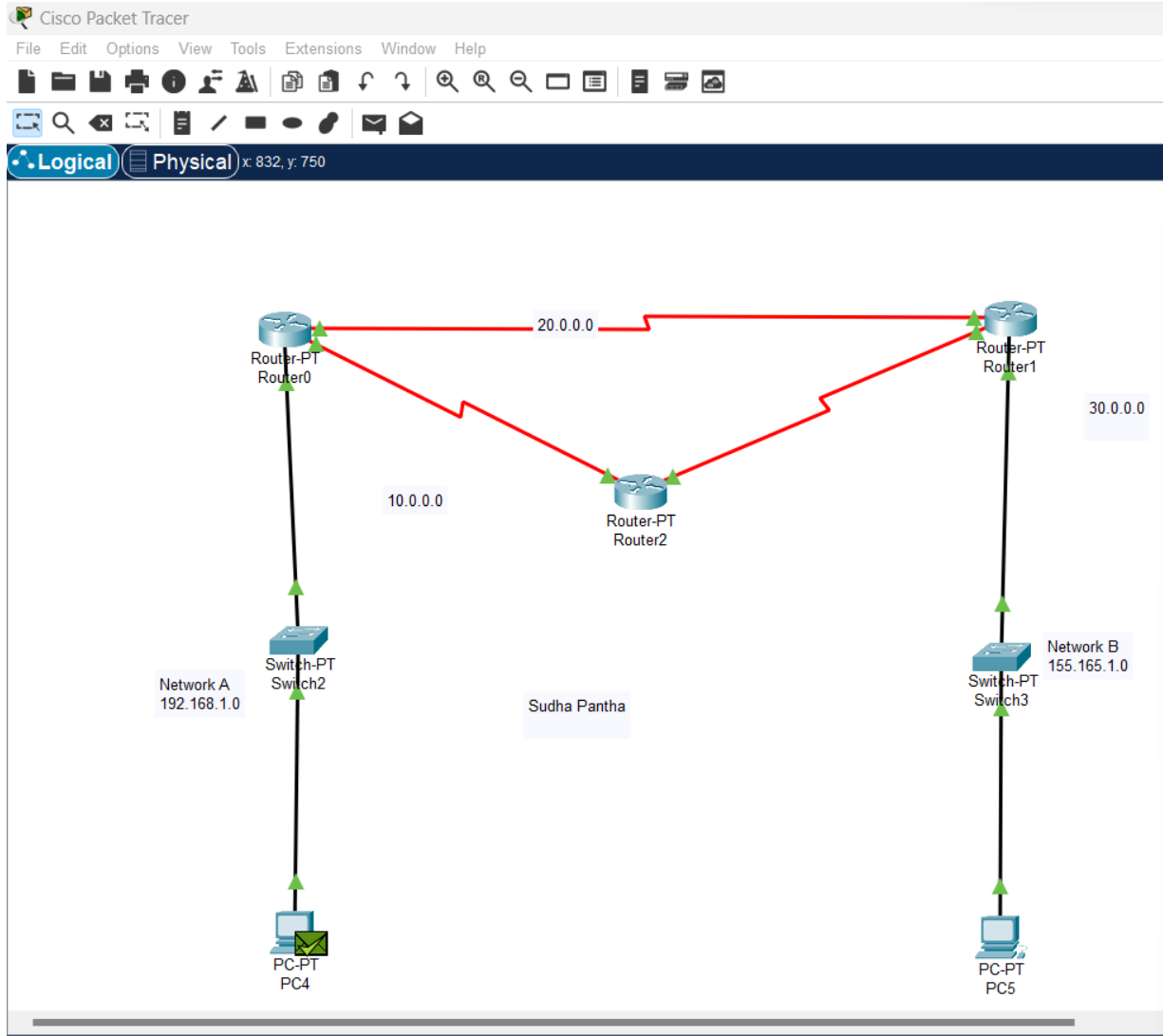


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Step 5: Simulation

- A simulation of the experiment is given below we are sending PDU from PC0 to PC1:



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Simulation Panel				
Event List				
Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC4	ICMP
	0.001	PC4	Switch2	ICMP
	0.002	Switch2	Router0	ICMP
	0.003	Router0	Router1	ICMP
	0.004	Router1	Switch3	ICMP
	0.005	Switch3	PC5	ICMP
	0.006	PC5	Switch3	ICMP
	0.007	Switch3	Router1	ICMP
	0.008	Router1	Router0	ICMP
	0.009	Router0	Switch2	ICMP
	0.010	Switch2	PC4	ICMP

Scenario 0

New

Delete

Toggle PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC4	PC5	ICMP		0.000	N	0	(edit)	(delete)

ENG IN

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Step 6: Verifying the network by pinging the IP address of any PC.

PC4

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time=2ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time=2ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 1ms
C:\>
```

ENG IN

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Name: Sudha Pantha



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:
Reply from 192.168.1.11: bytes=32 time=2ms TTL=128
Reply from 192.168.1.11: bytes=32 time=1ms TTL=128
Reply from 192.168.1.11: bytes=32 time=2ms TTL=128
Reply from 192.168.1.11: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 1ms

C:\>ping 155.165.1.11

Pinging 155.165.1.11 with 32 bytes of data:
Reply from 155.165.1.11: bytes=32 time=10ms TTL=126
Reply from 155.165.1.11: bytes=32 time=10ms TTL=126
Reply from 155.165.1.11: bytes=32 time=10ms TTL=126
Reply from 155.165.1.11: bytes=32 time=10ms TTL=126

Ping statistics for 155.165.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 10ms, Average = 10ms

C:\>
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 155.165.1.11

Pinging 155.165.1.11 with 32 bytes of data:
Reply from 155.165.1.11: bytes=32 time=2ms TTL=128
Reply from 155.165.1.11: bytes=32 time=2ms TTL=128
Reply from 155.165.1.11: bytes=32 time=4ms TTL=128
Reply from 155.165.1.11: bytes=32 time=1ms TTL=128

Ping statistics for 155.165.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 4ms, Average = 2ms

C:\>
```

Conclusion:

Hence, we configured OSPF protocol successfully verified the connection.

Name: Sudha Pantha



Lab Program Number: ...15...

Date: 2080-10-22

Title: To configure and understand about static routing using cisco packet tracer.

Theory:

Static routing is a routing protocol that helps to keep your network organized and to optimize routing performance. It enables the router to assign a specific path to each network segment and to keep track of network changes. This helps to improve network stability and continuity. This adds security because a single administrator can only authorize routing to particular networks.

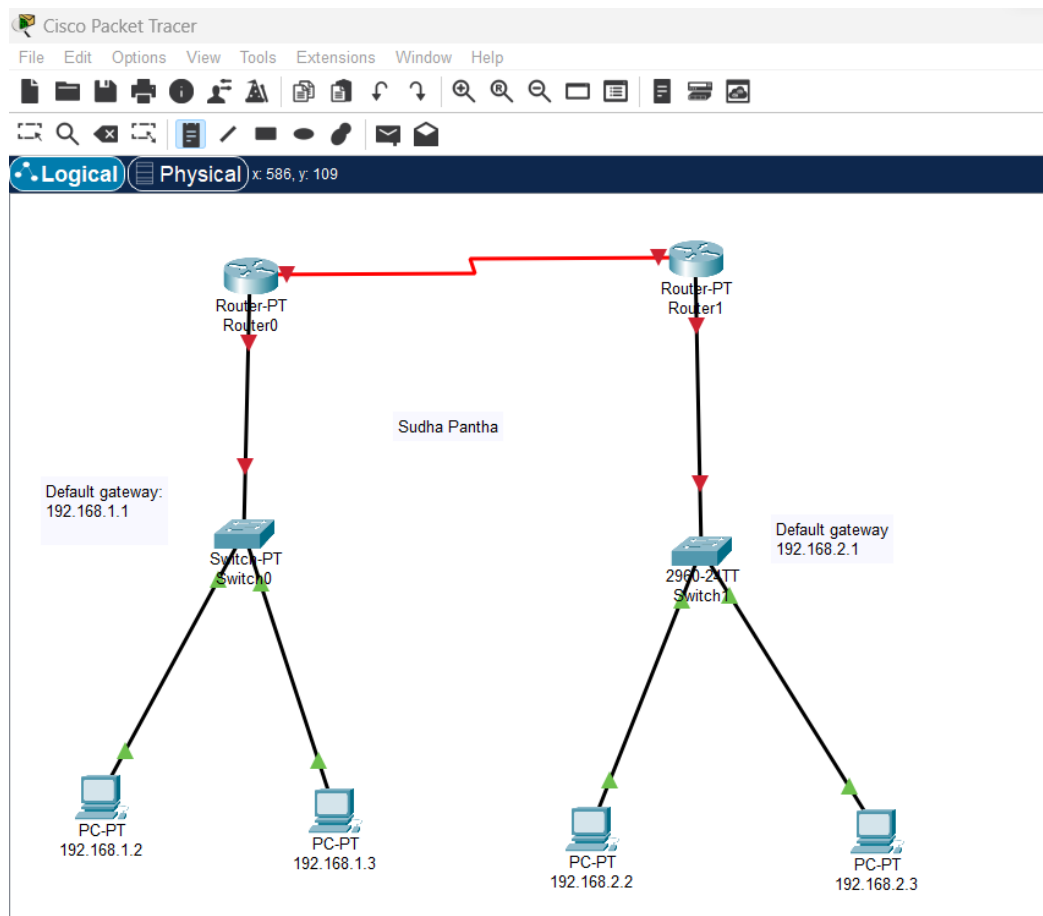
Procedure:

Step 1: First, open the cisco packet tracer desktop and select the devices given below:

S.No	Device	Model Name	Qty.
01	PC	PC	04
02	Switch	PT-Switch	02
03	Router	Router-PT	02

- In the “Devices” tab on the left, select “End Devices” and drag a “Generic” Pc or laptop to the workspace.
- Similarly, drag a “Switch” and “Router” to the workspace.
- Click on the “Connections” icon on the left-hand side of the screen.
- Choose the appropriate cable type usually, Ethernet connections use straight-through cables.
- Click on one device’s interface port and then click on the other devices interface to establish a connection.
- Then, create a network topology as shown below the image.

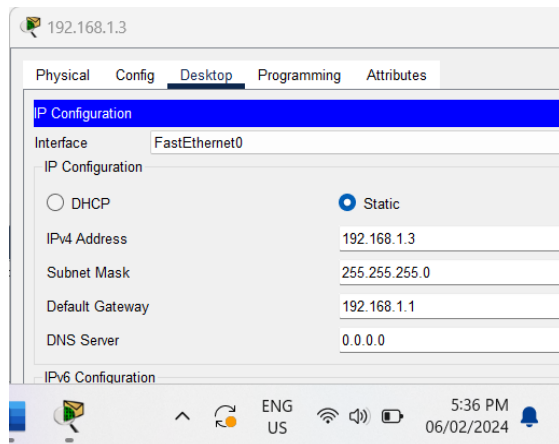
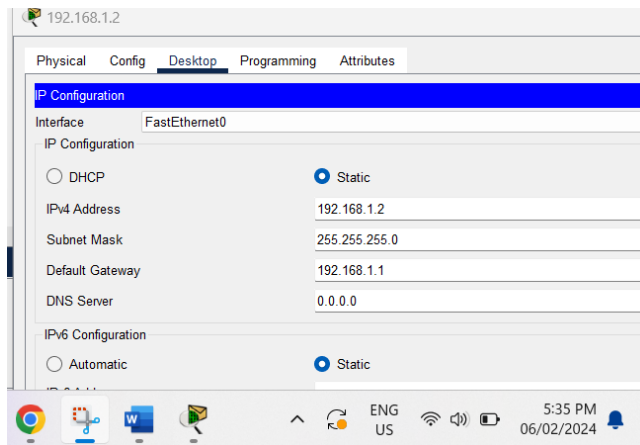
Name: Sudha Pantha



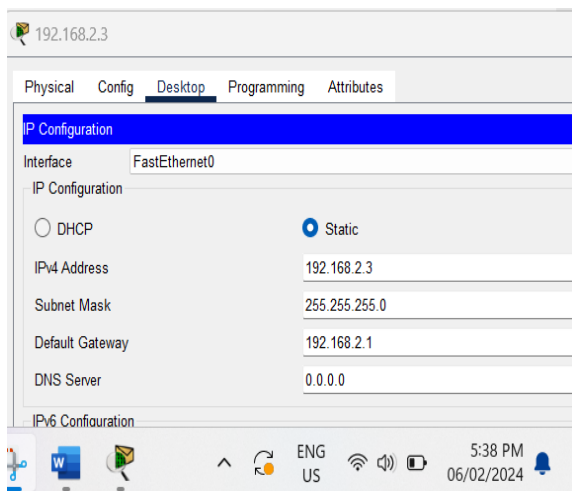
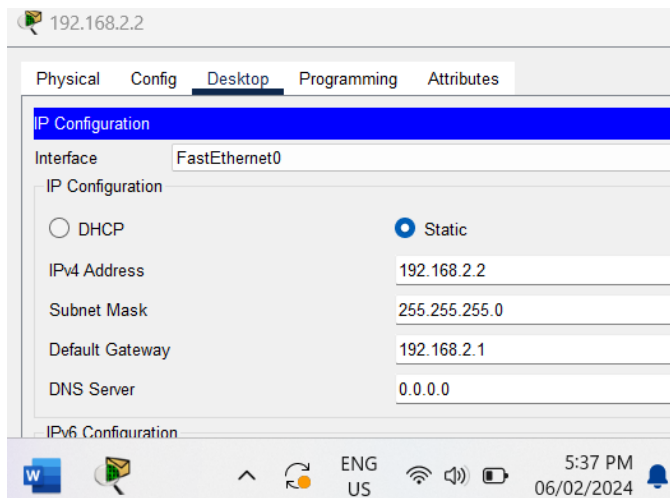
Step 2: Configure the PCs (hosts) with IPv4 address and Subnet Mask according to the IP addressing table given above.

- To assign an IP address in all PCs.
- Then, go to desktop and then IP configuration and there you will IPv4 configuration.
- Fill IPv4 address and subnet mask.

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Name: Sudha Pantha



Step 3: Configure router with IP address and Subnet

mask.

- To assign an IP address in router, click on router.
- Then, go to config and then Interfaces.
- Make sure to turn on the ports.
- Then, configure the IP address in Fast Ethernet and serial ports according to IP addressing Table.
- Fill IPv4 address and subnet mask for all router.

Name: Sudha Pantha



Router0

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

FastEthernet0/0

Port Status ☒ On

Bandwidth ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☐ Full Duplex ☒ Auto

MAC Address 00D0.D302.8682

IP Configuration

IPv4 Address 192.168.1.1

Subnet Mask 255.255.255.0

5:40 PM 06/02/2024

Router0

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

Serial2/0

Port Status ☒ On

Duplex ☐ Full Duplex

Clock Rate 64000

IP Configuration

IPv4 Address 11.0.0.1

Subnet Mask 255.0.0.0

Tx Ring Limit 10

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Name: Sudha Pantha



Router1

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

INTERFACE

- FastEthernet0/0**
- FastEthernet1/0
- Serial2/0
- Serial3/0
- FastEthernet4/0

FastEthernet0/0

Port Status ☒ On

Bandwidth ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☐ Full Duplex ☒ Auto

MAC Address 0005.5E00.9390

IP Configuration

IPv4 Address 192.168.2.1

Subnet Mask 255.255.255.0

Tx Ring Limit 10

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Router1

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

INTERFACE

- FastEthernet0/0
- FastEthernet1/0
- Serial2/0**

Serial2/0

Port Status ☒ On

Duplex ☐ Full Duplex

Clock Rate Not Set

IP Configuration

IPv4 Address 11.0.0.2

Subnet Mask 255.0.0.0

Tx Ring Limit 10

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Step 4: After configuring all of the devices we need to assign the routes to the routers.

- To assign static routes to the particular router
- First, click on router0 then Go to CLI.
- Then type the commands and Ip information given below.

```
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 192.168.1.0 255.255.255.0 11.0.0.1
Router(config)#
```

```
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 192.168.1.0 255.255.255.0 11.0.0.1
Router(config)#
```

Step 5: Verifying the network by pinging the IP address of any PC. We will use the ping command to do so.

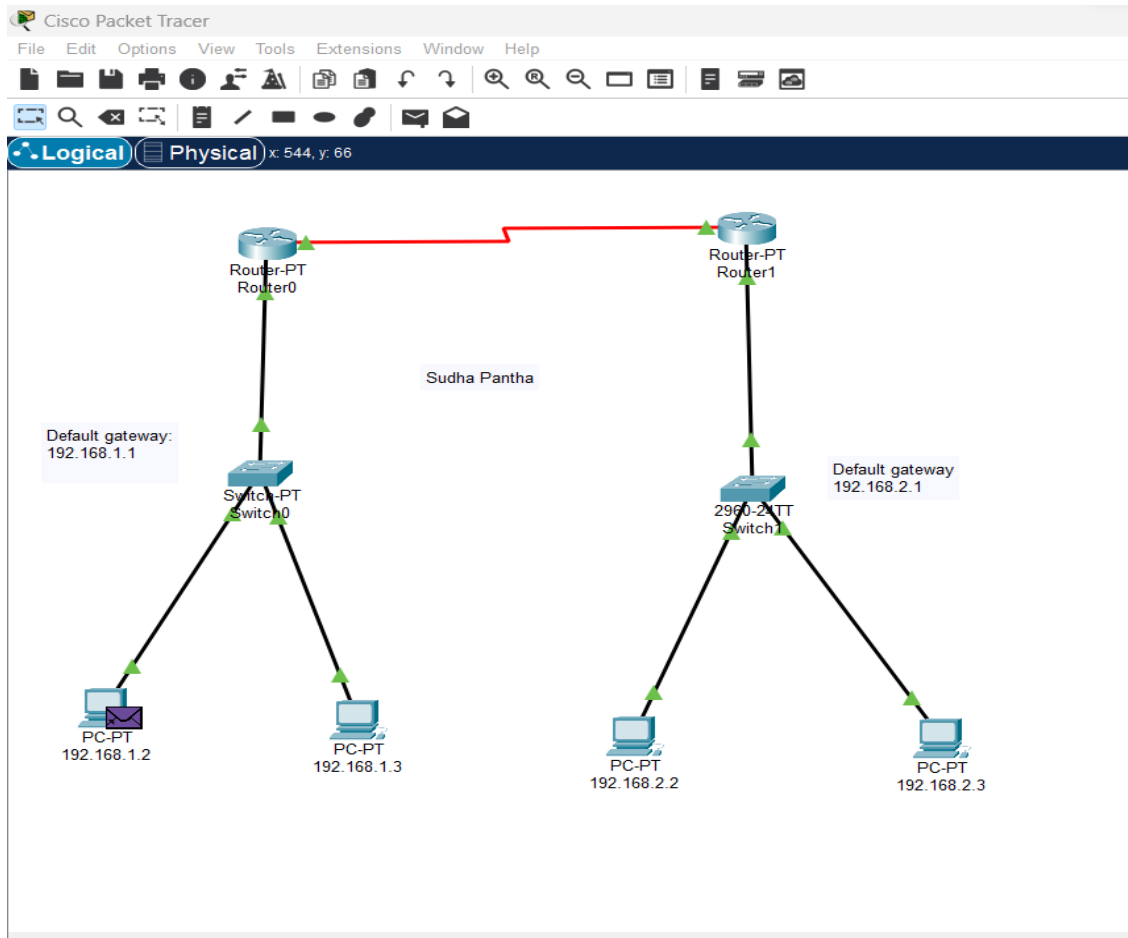
- First, click on PC1 then Go to the command prompt
- Then type ping <IP address of targeted node>
- As we can see in the below image we are getting replies which means the connection is working very fine

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Step 6: We are sending PDU from PC0 to PC1.

Name: Sudha Pantha



Name: Sudha Pantha



Simulation Panel

Root

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	192.168.1.2	ICMP
	0.004	--	192.168.1.2	ICMP
	0.005	192.168.1.2	Switch0	ICMP
	0.006	Switch0	192.168.1.3	ICMP
	0.007	192.168.1.3	Switch0	ICMP
	0.008	Switch0	192.168.1.2	ICMP

Reset Simulation ☒ Constant Delay

Captured to: 0.008 s

Play Controls

Event List Filters - Visible Events

ACL Filter, Bluetooth, CAPWAP, CDP, DHCPv6, DTP, EAPOL, EIGRPv6, FTP, H.323, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPSec, ISAKMP, IoT, IoT TCP, LACP, LLDP, Meraki, NDP, NETFLOW, NTP, OSPFv6, PAgP, POP3, PPP, PPPoE, PTP, RADIUS, REP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

Edit Filters Show All/None

Event List Realtime Simulation

icenario 0	Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
		Successful	192.1...	192.168.1.3	ICMP		0.000	N	0	(edit)	(delete)

Delete

U List Window

ENG 6:02 PM

Conclusion:

Hence, we configured static routing protocol successfully verified the connection.

Name: Sudha Pantha